Digital Health Policy and Programmes for Hospital Care in Vietnam: A Scoping Review

Duc Minh Tran, C Louise Thwaites, Jennifer Ilo Van Nuil, Jacob McKnight, An Phuoc Luu, Chris Paton, Phil Vietnam ICU Translational Applications Laboratory (VITAL)

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Duc Minh Tran[1] PMD; C Louise Thwaites[1, 2] BSc, MBBS, MD; Jennifer Ilo Van Nuil[1, 2] BA, MA, PhD; Jacob McKnight[3] BEng, MSc, DPhil; An Phuoc Luu[1] MPH; Chris Paton[3, 4] BMBS, BMedSci, MBA; Phil Vietnam ICU Translational Applications Laboratory (VITAL)[5]

[1]Oxford University Clinical Research Unit Ho Chi Minh City VN
[2]Centre for Tropical Medicine and Global Health University of Oxford Oxford GB
[3]Nuffield Department of Medicine University of Oxford Oxford GB
[4]Department of Information Science University of Otago Otago NZ
[5]See Acknowledgments Melbourne AU

Corresponding Author:
Duc Minh Tran PMD
Oxford University Clinical Research Unit
764 Vo Van Kiet, Ward 1, District 5
Ho Chi Minh City
VN

Abstract

Background: A host of emergent technologies have the potential to improve hospital care in low-and-middle income countries (LMICs) such as Vietnam. It has been suggested that wearable monitors and artificial intelligence (AI)-based decision support systems could be integrated with hospital-based digital health systems such as electronic health records (EHRs) to provide higher level care at relatively low cost. However, the appropriate and sustainable application of these innovations in LMICs requires an understanding of the local government's requirements and regulations such as technology specifications, cybersecurity, data sharing, and interoperability.

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Methods: We conducted a scoping review using a modification of the Preferred Reporting Items for Systematic reviews and Meta-Analyses extension for Scoping Reviews (PRISMA-ScR) guidelines. PubMed and Web of Science were searched for Academic publications, and thuvienphapluat.vn, a proprietary database of Vietnamese government documents, and the Vietnam Electronic Health Administration website, were searched for government documents. Google Scholar and Google Search were used for snowballing searches. The sources were assessed against a pre-defined eligibility criteria through title, abstract, and full text screening. Relevant information from the included sources was charted and summarized. The review steps were primarily undertaken by one researcher and subsequently reviewed by another researcher during each step.

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Conclusions: Research about hospital-based digital health systems in Vietnam is very limited with implementation studies particularly lacking. Government regulations and guidance for HIT in healthcare organizations have been released with increasing frequency since 2013, targeting a variety of information systems such as EMRs, EHRs, and LISs. In general, these policies were focused on the basic specifications and standards that the digital health systems need to meet. More research is needed in the future to guide the implementation of digital healthcare systems in the Vietnam hospital settings.

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Review

Authors information

#	**Full Name** ([Given] [Middle] (optional) [Family])	**Author degree**	**Affiliations** (Please list all of your affiliations)	**Email**	**ORCID**
1	Duc Minh Tran	PMD	Oxford University Clinical Research Unit, District 5, Ho Chi Minh City, Vietnam.	ductm@oucru.org	0000-0002-1151-5988
2	C Louise Thwaites	BSc MBBS MD	Oxford University Clinical Research Unit, District 5, Ho Chi Minh City, Vietnam. Centre for Tropical Medicine and Global Health, University of Oxford.	lthwaites@oucru.org	0000-0002-4666-9813
3	Jennifer Ilo Van Nuil	BA MA PHD	Oxford University Clinical Research Unit, District 5, Ho Chi Minh City, Vietnam. Centre for Tropical Medicine and Global Health, University of Oxford.	jvannuil@oucru.org	0000-0002-5167-5505
4	Jacob McKnight	BEng(hons), MSc., DPhil	Nuffield Department of Medicine, University of Oxford.	jacob.mcknight@ndm.ox.ac.uk	0000-0003-1340-2618
5	An Phuoc Luu	MPH	Oxford University Clinical Research Unit, District 5, Ho Chi Minh City, Vietnam.	anlp@oucru.org	0000-0002-5651-1091
6	Chris Paton	BMBS BMedSci MBA	Nuffield Department of Medicine, University of Oxford. Department of Information Science, University of Otago.	chris.paton@ndm.ox.ac.uk	0000-0001-6952-9621
7	Vietnam ICU Translational Applications Laboratory (VITAL) investigators	**OUCRU inclusive authorship list in Vietnam** (alphabetic order by surname): Sayem Ahmed, Dang Trung Kien, Doan Bui Xuan Thy, Dong Huu Khanh Trinh, Joseph Donovan, Du Hong Duc, Ronald Geskus, Ho Bich Hai, Ho Quang Chanh, Ho Van Hien, Hoang Minh Tu Van, Huynh Trung Trieu, Evelyne Kestelyn, Lam Minh Yen, Le Dinh Van Khoa, Le Thanh Phuong, Luu Phuoc An, Angela Mcbride, Nguyen Lam Vuong, Nguyen Quang Huy, Nguyen Than Ha Quyen, Nguyen Thanh Ngoc, Nguyen Thi Giang, Nguyen Thi Le Thanh, Nguyen Thi Phuong Dung, Ninh Thi Thanh Van, Pham Thi Lieu, Phan Nguyen Quoc Khanh, Phung Khanh Lam, Phung Tran Huy Nhat, Guy Thwaites, Louise Thwaites, Tran Minh Duc, Trinh Manh Hung, Hugo Turner, Jennifer Ilo Van Nuil, Sophie Yacoub **Hospital for Tropical Diseases, Ho Chi Minh City** (alphabetic order by surname): Cao Thi Tam, Duong Bich Thuy, Ha Thi Hai Duong, Ho Dang Trung Nghia, Le Buu Chau, Le Ngoc Minh Thu, Le Thi Mai Thao, Luong Thi Hue Tai, Nguyen Hoan Phu, Nguyen Quoc Viet, Nguyen Thanh Nguyen, Nguyen Thanh Phong, Nguyen Thi Kim Anh, Nguyen Van Hao, Nguyen Van Thanh Duoc, Nguyen Van Vinh Chau, Pham Kieu Nguyet Oanh, Phan Thi Hong Van, Phan Tu Qui, Phan Vinh Tho, Truong Thi Phuong Thao **University of Oxford** (alphabetic order by surname): David Clifton, Mike English, Shadi Ghiasi, Heloise Greeff, Jannis Hagenah, Ping Lu, Jacob			

		McKnight, Chris Paton, Tingting Zhu **Imperial College London** (alphabetic order by surname): Pantellis Georgiou, Bernard Hernandez Perez, Kerri Hill-Cawthorne, Alison Holmes, Stefan Karolcik, Damien Ming, Nicolas Moser, Jesus Rodriguez Manzano **King's College London** (alphabetic order by surname): Alberto Gomez, Hamideh Kerdegari, Marc Modat, Reza Razavi **ETH Zurich** (alphabetic order by surname): Abhilash Guru Dutt, Walter Karlen, Michaela Verling, Elias Wicki **The University of Melbourne** (alphabetic order by surname): Linda Denehy, Thomas Rollinson

Corresponding author:
Duc Minh Tran;
Oxford University Clinical Research Unit, 764 Vo Van Kiet, Ward 1, District 5, Ho Chi Minh City, Vietnam;
Phone: +84356574593
Email: ductm@oucru.org

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Keywords: digital health; eHealth; policy; Vietnam; hospital care.

Introduction

Digital health systems such as electronic health records (EHRs) and patient administration systems used in hospitals in high income countries (HICs) have been adopted with the dual aims of increasing the quality of patient care and improving hospital finances through cost-reductions and new revenue streams. These systems are commonly introduced in response to major government initiatives, often with significant public funding[1]. Despite major challenges and high-profile failures[2], HICs have now reached the point where secondary use of data from digital health systems can, in some cases, enable hospitals and healthcare systems to become ‘learning health systems’, using routinely collected data to facilitate research and quality improvement[3]. In more recent years, data from hospital-based digital health systems have been used to develop and implement innovative artificial intelligence (AI) systems for monitoring patients and providing clinical decision support (CDS) to healthcare providers[4].

While the adoption of digital health systems in low-and-middle income countries (LMICs) such as Vietnam has largely only taken place within the last decade, these solutions have the potential to support the development of universal health coverage and projects working towards addressing the sustainable development goals (SDGs)[5,6]. However, in resource-constrained settings, new healthcare information technologies are often implemented with insufficient funding, infrastructure, regulations, and computer literacy of the staff who will be using them[7–9]. These challenges may be able to be mitigated through the use of open-source software[8,10], mobile technologies[11] and cloud-based data infrastructure[12]. The adoption of new policies and standards (such as HL7 FHIR[13]), may also enable simpler and more effective methods for health information exchange than was available to HICs in previous years[14].

Recent initiatives in Vietnam and other LMICs have sought to exploit the potential of digital technologies such as machine learning and low-cost wearable devices in improving critical care at an affordable cost[15,16]. For these innovations to attain scalability and sustainability, their research and development needs to consider the technical infrastructure and the regulatory frameworks that govern local technology adoption[5,6,15,16]. The Vietnamese government has recognized the role that digital health technologies can play in improving health care and optimising administration processes[17], and, along with building national health databases such as the national EHR system, new digital health policies and guidance have been promulgated by the Vietnam Ministry of Health (MoH)[18,19]. Awareness of the government’s current regulations and future directions for digital health, as well as the local research evidence in this area, is important for the introduction of these emergent technologies in Vietnamese hospitals. This scoping review aims to map and summarize the academic literature and government policies in the field of hospital-based digital health systems in Vietnam. A scoping review methodology was chosen as it allowed us to effectively explore and summarize information from a wide range of sources given the rapidly evolving nature of digital health in Vietnam[20,21].

Methods

We conducted this review using a modification of the “PRISMA Extension for Scoping Reviews (PRISMA-ScR): Checklist and Explanation” guidelines[22]. Details of each step are as follows.

Eligibility criteria

Inclusion criteria

- Policies and guidance documents from the government that regulate and guide the adoption of digital health systems in Vietnam's public hospitals. Academic publications that address digital health systems in Vietnam's hospitals.
- Policies and guidance documents that are functioning or to be mandated.
- Documents written in Vietnamese or English.

Exclusion criteria

- Policies and guidelines that have been replaced by a newer version.
- Policies and studies that only examine technical aspects of the digital solution without discussing its adoption and implementation in clinical settings.
- Standalone apps or digital solutions that are implemented in the hospital settings but not linked with a particular hospital-based information system such as hospital information management systems (HIMSs) and electronic medical records (EMRs).

Information sources

Searches from PubMed, Web of Science, and Google Scholar were conducted to identify academic literature. A structured search string (see Multimedia Appendix 1) was built for the search on PubMed and Web of Science, and additional relevant publications identified during the full text screening were retrieved and screened using Google Scholar.

A search of Vietnamese policies was conducted on the Thư Viện Pháp Luật database[23]. Ten search queries were built and executed separately (see Multimedia Appendix 1). All document titles resulting from each of the search queries were extracted and compiled in an Excel spreadsheet, from which all duplicates were removed. The MoH Electronic Health Administration's website was used to search for relevant documents that had not been found in the structured database search. Follow-up visits to the website were done throughout the study and at the end of the data charting phase to identify newly-released documents. Relevant documents that were referred to in the reviewed text but not found by the two former search strategies were searched for using Google.

No restriction on publication type and publication year was set.

Data charting process

A data charting form was developed based on the research objectives and guidelines from the PRISMA-ScR guidelines. A draft version of the data charting form was piloted during the full text screening, from which discussions were made to modify and finalise the charting form.

All the charting tasks were conducted by one researcher (DT) using a Microsoft Excel spreadsheet. The charted data were then reviewed by another researcher (CP) and follow-up discussions were arranged to resolve any disagreements between the two investigators.

For academic literature, the charting form collected information about publication year, methods, study population or source of information reviewed (if the publication is a review), study scope, the digital health domain being studied, the context that the domain was investigating, and key findings. The charting form of government documents sought to extract information about the domain of intervention, policy enactment time, the intended purpose of the policy, and the ministry that mandated the document.

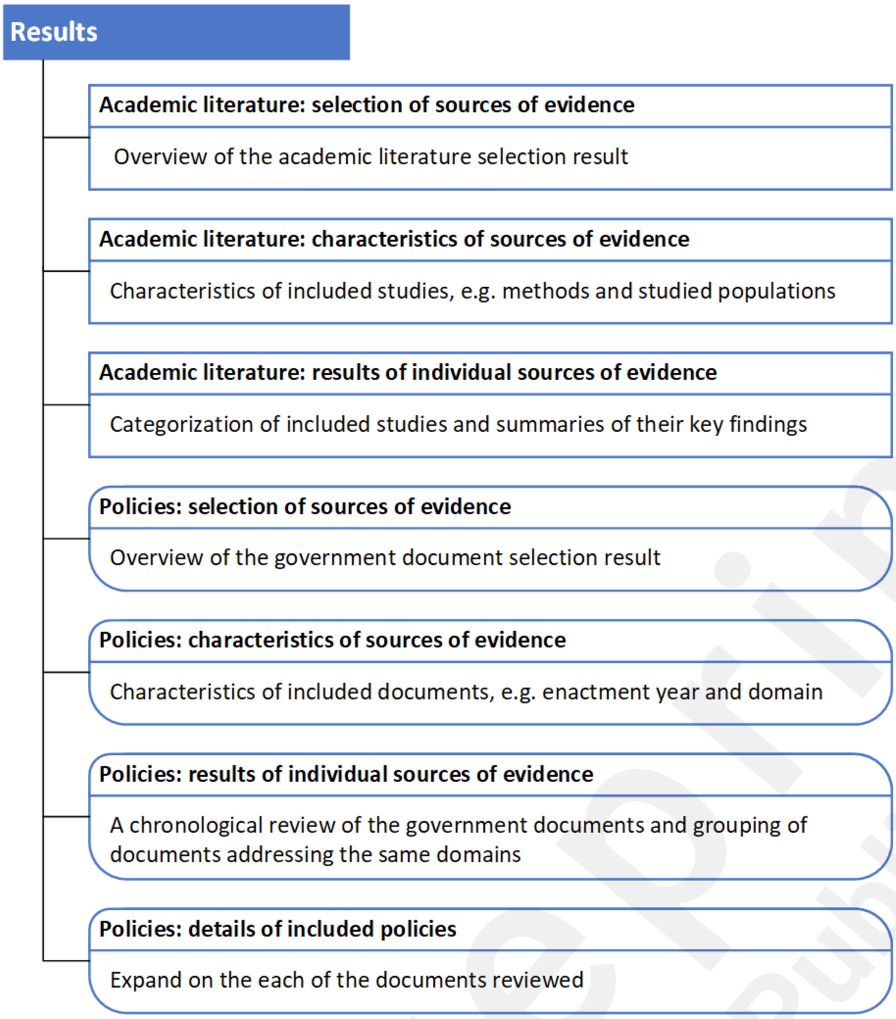
Selection of sources of evidence

Academic publications were assessed through three steps including title, abstract and full text screening. Studies that did not meet the eligibility criteria in each step were ruled out from the screening list. Government policies and guidance were selected through a two-stage process. In the first screening round, the title, introduction and 'scope and regulated entities' section of each document was screened to decide their eligibility. Documents which passed the first round then had their full texts screened in the second screening round. Selection was based on the eligibility criteria described above. Any uncertainty during the selection process was discussed between the authors for a final agreement.

Results

From an initial search results of 2033 academic publications and 266 government documents, 11 academic studies and 20 government documents were included in the review. In the following section, we organized the results in two major parts, academic literature and policies. Each part featured the source selection process, characteristics of selected sources, and the results of individual selected sources. Figure 1 depicts the layout of the Results section.

Figure 1. Layout of the Results section

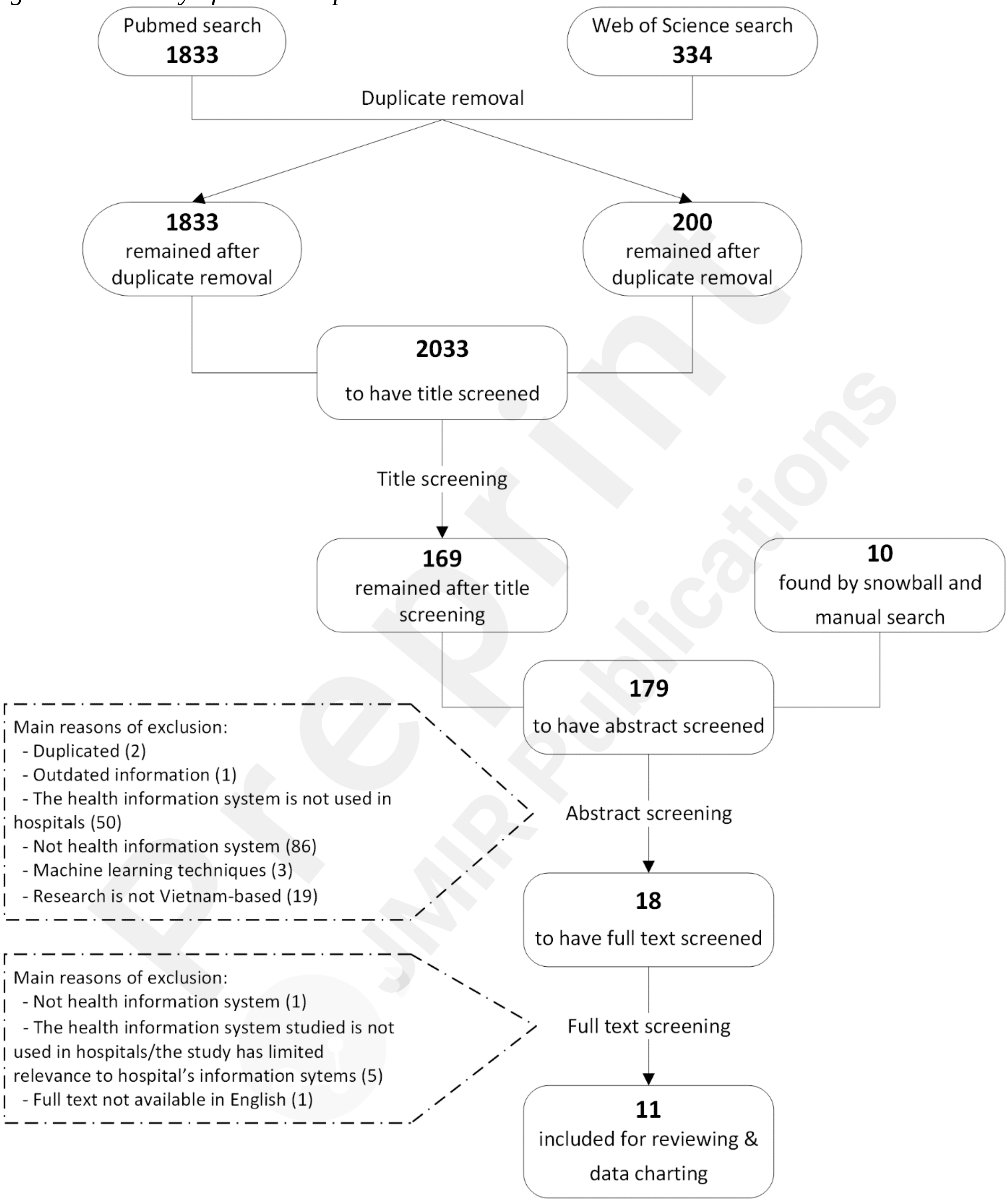


[a]The branches represent the subsections in the Results section and the information included herein. Square boxes and round boxes denote the "academic literature" and "policies" subsections respectively.

Academic literature: selection of sources of evidence

The PubMed and Web of Science searches of the academic literature returned 1833 and 334 articles respectively. After removal of duplicates, title screening was conducted on 2033 articles (1833 from PubMed and 200 from Web of Science), resulting in 169 marked "considered" for abstract screening. Titles were excluded if they did not imply any relation to the use of information technology (IT) in health care. In addition, the snowballing and manual search found an addition of 10 studies to make a total of 179 abstracts to be screened. The abstract screening ruled out 161 articles that did not meet the eligibility criteria and identified 18 articles eligible for the full text review. The main reasons for exclusion during abstract screening were duplicated article (2), the reviewed information was outdated to the current Vietnam's digital health situation (1), the interventions studied were neither health information systems (86) nor hospital's health information systems (50), the research was about machine learning techniques (3), and that the research was not conducted in Vietnam (19). Out of the full texts screened, 11 articles were included in the final review. The full text screening found 7 studies not eligible for the review, of which one did not examine health information systems, five studies investigated health information systems that were not implemented in hospitals or their findings were of limited relevance to hospital's information systems, and one study's full text was not available in English (see Figure 2).

Figure 2. Summary of academic publication selection



Academic literature: characteristics of sources of evidence

Characteristics of each study including title, authors, publication year, methods, studied population/ source of information reviewed, and study's scope was summarized in Table 1.

Table 1. Summary of included studies

Study title	Author	Methods	Studied population	Study's scope

	s & publication year		or source of information reviewed	
Design of Laboratory Information System for healthcare in Vietnam BK-LIS[32]	Vu Duy Hai et al. 2010	Case study	Common lab test results in Vietnam's hospitals	Described how BK-LIS, a laboratory information system, was designed and developed to support the lab activities in Vietnam's hospitals.
A Design of Renal Dataflow Control and Patient Record Management System for Renal Department Environment in Vietnam[33]	Vu Duy Hai et al. 2010	Case study	Haemodialysis systems in Bach Mai hospital and E Hospital, two central-level hospitals in Vietnam	A case study of the development of BK-HDmanager, an IT solution that centrally manages the haemodialysis system in hospital.
Automatic Retrieving Data from Medical Equipment to Create Electronic Medical Records for an e-Hospital Model in Vietnam[31]	Vu Duy Hai et al. 2011	Case study	Electronic medical equipment commonly used in Vietnamese hospitals	Demonstrated technical solutions to automatically retrieve data from medical devices. Types of data include images and video data, lab test result data, and waveform data.
Towards VNUMED for Healthcare Research Activities in Vietnam[30]	Chau Vo et al. 2019	Case study	Electronic medical records (EMRs) from hospitals in Vietnam	Introduction of VNUMED, an intermediate database that gathers data from EMRs to support healthcare research, and related challenges for its development in Vietnam.
EMRVisualization for Patient Progress Tracking: Interface Design and	Chau Vo et al. 2019	Case study	Gastroenterologists and EMR data in Thong Nhat central-level hospital	Described the development and testing of EMRVisualization, a visualization tool for patient progress tracking, using data from the hospital's EMR system.

System Implementation [30]				
Strategic challenges facing user- and patient-centred e-health in Vietnam[27]	Nguyen et al. 2012	Review	IT use in health care in Vietnam before 2012	Review of IT applications in Vietnam's health sector before 2012. Challenges in implementing patient-centred eHealth in Vietnam were discussed.
English-based Pediatric Emergency Medicine Software Improves Physician Test Performance on Common Pediatric Emergencies: A Multicentre Study in Vietnam[26]	Lin et al. 2013	Multicentre, prospective, pretest-posttest study	203 physicians from 11 hospitals across Vietnam	PEMSoft, a clinical decision support system, was tested against physicians' performance on a multiple-choice exam.
Towards an electronic health record system in Vietnam: a core readiness assessment. [28]	Hocwarter et al. 2014	Document analysis, participant observation and in-depth interview	Participant observation and document analysis was conducted in a department of a top-level hospital in Vietnam; in-depth interview e with an MoH expert.	Investigation of the Vietnamese health system's core readiness for electronic health record implementation.
Open-source LIMS in Vietnam: The path toward sustainability and host country ownership[24]	Landgraf et al. 2016	Reviewing the program reports	Reports from a national laboratory information management system (LIMS) project utilizing an open-source LIMS from 2008 to 2016	Described the building and scale-up of a national LIMS program for clinical and public health laboratories in Vietnam. Outcomes of the program and the lessons learned were discussed. A model for sustainability that could be applied to diverse laboratory programs was proposed.
Electronic Health Record Readiness	Nguyen et al. 2018	Cross-sectional study	Thai Binh provincial hospital	The study assessed the readiness for EHR implementation in Thai Binh hospital, Vietnam. The four main

Assessment in Thai Binh Hospital, Vietnam[29]		using a scoring tool		components of readiness included core readiness, technological readiness, learning readiness, and societal readiness.
Artificial Intelligence vs. Natural Stupidity: Evaluating AI readiness for the Vietnamese Medical Information System[25]	Vuong el al. 2019	Non-systematic review	Literature about artificial intelligence (AI) in medicine worldwide, literature about eHealth in Vietnam, and the Joint Annual Health Review of Vietnam from 2012 to 2016	An overview of AI research and applications in medicine worldwide, proposing a framework to evaluate AI readiness. Assessment of AI readiness in Vietnam's healthcare sector using the proposed framework.

Academic literature: results of individual sources of evidence

The digital health domain that each study addressed, the context that the domain was considered, year of data collection or reviewed evidence, and key findings of each study are presented in Table 2.

Among the 11 publications reviewed, 5 reported the development and testing of engineering solutions to gather, manage, or visualize data from the medical devices or information systems that were being employed in the chosen hospitals. The 3 studies published in 2010 and 2011 described solutions to retrieve and manage data from laboratory devices or haemodialysis systems, whereas the 2 studies published in 2019 involved EMR-based solutions.

The other 6 articles addressed aspects of digital health implementation in Vietnam's healthcare system. Among these, none examined a specific clinical information system in the hospital setting. EHR implementation was discussed in the two readiness assessment studies, in which one interviewed an MoH staff while the other study surveyed healthcare workers in a provincial hospital. These stakeholders recognized the benefits offered by EHRs and expressed positive attitudes towards EHR adoption. However, there was a lack of IT infrastructure, basic IT utilization among the hospital staff such as regular communication via email, and IT training capacity in place for EHR implementation. Lin and colleagues evaluated physicians' performance on a multiple-choice exam with the support from a CDS system[26]. The participants improved their exam performance when using the CDS software compared to not using the CDS. However, the software was only used for testing purposes and not implemented in the studied hospitals. The national LIMS program published by Landgraf and colleagues was largely utilized in public health laboratories rather than integrated in a hospital-based system[24]. The two reviews from Nguyen[27] and Vuong[25] provided an overview of how healthcare facilities in Vietnam had implemented IT applications. Specifically, barriers for the development of patient-centred eHealth and AI in medicine in Vietnam were summarized, including the lack of national health databases, unstandardized data collection, and the paper-based workflows. Importantly, most of the data analyzed in these 6 studies were collected in 2016 or earlier. The CDS performance and the EHR readiness assessments were conducted in the period from 2010 to 2014. Landgraf's paper was based on program reports from 2008 to 2016, whilst Nguyen's review provided insights into Vietnam eHealth prior to 2012. Discussions in Vuong's review was significantly made upon the analysis of the MoH annual reports from

2012 to 2016.

Table 2. Summary of the digital health domains and findings from the academic publications

Study	**Digital health domain**	**In what context that the intervention was considered?**	**Year of data collection/reviewed evidence[a]**	**Summary of findings relevant to digital health systems in Vietnam's hospitals**
Vu Duy Hai et al. 2010[32]	Laboratory information system	Common lab test results in Vietnam's hospitals	2010 or earlier[b]	Presented how a laboratory information system was designed to solve the paper-based lab result management.
Vu Duy Hai et al. 2010[33]	Haemodialysis management system	Haemodialysis systems in Vietnam's hospitals	2010 or earlier[b]	A central management system for haemodialysis machines in two of Vietnam's hospitals was designed and tested.
Vu Duy Hai et al. 2011[31]	Data acquisition from medical devices	Electronic medical equipment commonly used in Vietnamese hospitals	2011 or earlier[b]	An engineering solution to automatically retrieve data from medical devices such as ultrasound, ECG, and laboratory devices for personal health records.
Chau Vo et al. 2019[30]	A database of electronic medical record (EMR) data	EMRs from hospitals in Vietnam	After 2019	The lack of standardized EMR use among Vietnamese hospitals posed a challenge for data gathering and research. VNUMED is a database that aims to collect and standardize data from different EMR systems.
Chau Vo et al. 2019[34]	Data visualization for EMRs	Data in EMR from a Vietnamese hospital	Between 2015 and 2019[b]	A data visualization application to track patient progress based on data collected from the local EMR system was developed and tested by the gerontologists as the end-users. The testing results showed positive feedback from the end-users regarding usability. The application was being updated for larger-scale testing.
Nguyen et al. [.2012[27]	Readiness for patient-centred eHealth	Health care in Vietnam before 2012	Before 2012	Provided an overview of IT applications in Vietnam's healthcare system from its beginning until 2011.
Lin et al. [.2013[2	Clinical decision	Physicians in hospitals	2010 to 2011	The study provided evidence that CDS technologies can improve physicians'

6]	support (CDS)			medical knowledge in the context of Vietnamese hospitals.
Hocwarter et al.[. 2014[28]	Electronic health record (EHR) readiness	Hospitals in Vietnam	2013	Provided evidence on the core readiness to a national EHR including: • Identification of needs for future changes that will be addressed by the EHR system • Challenges posed by the status quo that demanded an EHR system: the existing system of medical records; quality of the existing record keeping practice; the numbering system for medical records; patient identification methods in medical records; use of daily admissions and discharge lists; medical record archiving after patient discharge; medical record preservation when in archive; practice of ICD-10 and use of ICD-10 in reality. • Planning for the new EHR project by the Ministry of Health Integration of technology: integration with the current services; plan to integrate the EHR system with the existing hospital information systems; use of health informatics standards; the use of defined interfaces or gateways in data exchange.
Landgraf et al. 2016[24]	Laboratory information system	Mainly district health centres and public health laboratories in provinces with a high prevalence of HIV	2008 to 2016	Described the development, deployment, and operation of a national laboratory information management system (LIMS) project utilizing an open-source LIMS . Proposed factors for the sustainability of a health information system in VN: (1) selection of appropriate technology, (2) capacity building and knowledge transfer, (3) financial viability, (4) leadership and management, (5) alignment with national health strategies.
Nguyen et al.	EHR readiness	A provincial hospital in	2013 to 2014	Provided evidence on EHR readiness in a Vietnamese provincial hospital,

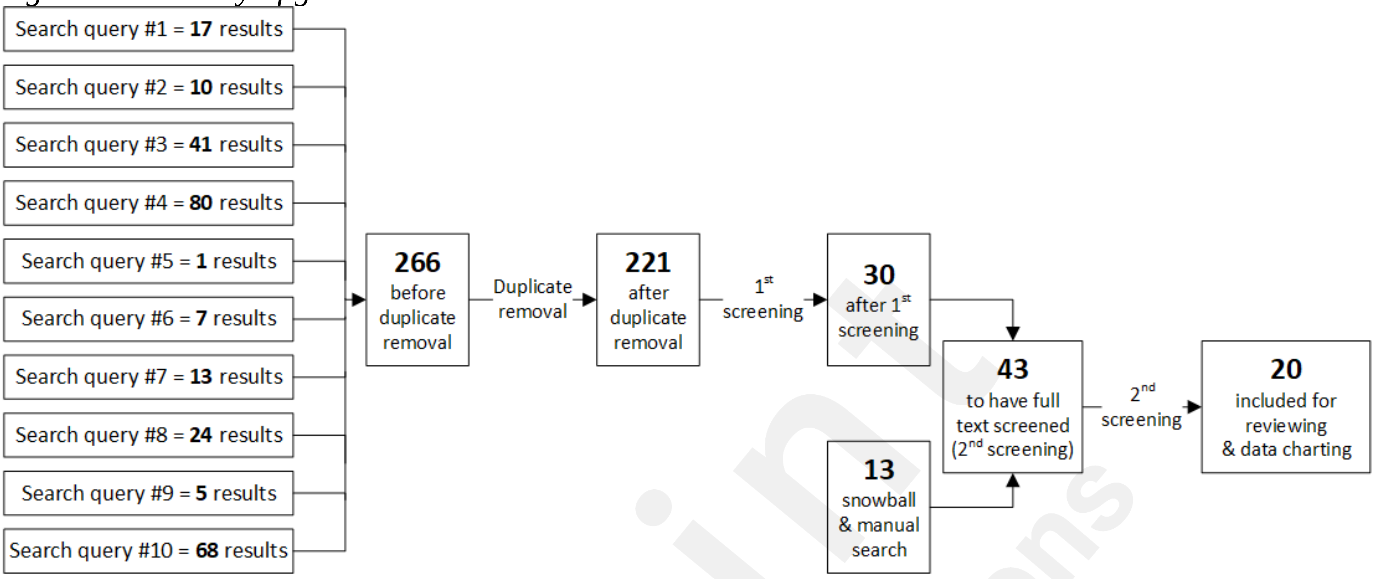
2018[29]		Vietnam		including: • Core readiness: needs for change, planning, suitability of infrastructure, and integration of new technology with the existing services in the hospital. • Technological readiness: need for IT adoption and IT infrastructure's capability to implement an EHR. • Learning readiness: IT training for hospital and implementation staff. Societal readiness: electronic communication with other organizations, data exchange between organizations, and sociocultural elements between health workers and patients.
Vuong el al. 2019[25]	Readiness for artificial intelligence (AI) in medicine	Health care of Vietnam	2012 to 2018 (mainly 2012 to 2016)	Readiness for AI in medicine in Vietnam was assessed based on 3 factors: financial support, technological, and socio-political. In general, while AIM research and political commitment was somewhat promising, the technical factor was seen as weak and inadequate.

[a]The period when primary data is collected or reviewed sources are published.
[b]Based on published year and reported grant period.

Policies: selection of sources of evidence

Ten search queries on the *thuvienphapluat.vn* database returned a total of 266 results. After removing duplicate titles, there were 221 results remaining in the 1st screening round. Thirty documents were selected from this screening phase with 191 documents excluded for the following reasons: unrelated to hospital information system adoption (157), limited relevance to hospital information system adoption (13), not a policy or guidance document (5), expired (13), and documents whose contents are covered in an included policy (3) (see Figure 3). Manual scanning on the MoH's Electronic Health Administration website and snowballing from the reviewed documents helped identify 13 potential policies and guidance documents. A total of 43 documents had their full texts screened, which yielded 20 documents included for review and data charting.

Figure 3. Summary of government document selection.



Policies: characteristics of sources of evidence

Table 3 presents an overview of the policies and guidance documents included. Characteristics of the documents were charted, including the enactment time points, policy IDs and titles, the digital health domains that the policies concerned, and the ministries that mandated the policies. Due to the lengthy original titles, shorter denoted titles were created for easier reference of the documents in this paper.

Table 3. Summary of policy documents

Valid from	Policy ID and title	Denoted title	Domain	Ministry
Dec 2006	Decision 5573/QD-BYT year 2006 on Guideline for hospital information management systems [35]	The HIMS Guidance	Hospital information management system	Ministry of Health (MoH)
Jun 2013	Decision 2035/QD-BYT year 2013 on Terminology systems and data exchange standards recommended for health IT[36]	The Recommended Standards for HIT	Standards for hospital information systems	MoH
Oct 2014	Decision 4159/QD-BYT year 2014 on Guidance on ensuring security of electronic health data in the health sector[37]	The Cybersecurity Guidance	Cybersecurity in health organizations	MoH
Mar 2015 or Jan 2017[a]	Circular 53/2014/TT-BYT on Required conditions for provision of health IT activities[38]	The Required Conditions for HIT	Conditions for provision of health IT	MoH
Oct 2015	Decision 4495/QD-BYT year 2015 on Guideline for developing local information safety and security rules in health facilities[39]	The Guidance for Local Cybersecurity Policy	Cybersecurity in health organizations	MoH
Oct 2015	Decision 4494/QD-BYT year 2015	The Guidance	Cybersecurit	MoH

	on Response procedures for information safety and security issues in the health sector[40]	for Cybersecurity Response	y in health organizations	
Sep 2016	Decision 5004/QD-BYT year 2016 on The architectural framework of the social health insurance information system[41]	The Social Health Insurance EAF	Electronic health insurance claim	MoH
Jun 2016	Decision 917/QD-BHXH year 2016 on Announcement of the health insurance portal version 2[42]	The Insurance Portal V2 Guidance	Electronic health insurance claim	Vietnam Social Security
Sep 2017	Decision 4210/QD-BYT year 2017 on Requirements for standard and format of output data used in management, assessment and reimbursement of insurance-paid health care expenses[43]	The Claim Standardization Guidance	Electronic health insurance claim	MoH
Aug 2017	Decision 3725/QD-BYT year 2017 on Guidelines for functionalities, interoperability, infrastructure and human resources for establishing and implementing Laboratory Information Systems at healthcare facilities[44]	The LIS Guidance	Laboratory information system	MoH
Feb 2018	Circular 54/2017/TT-BYT on Assessment Criteria for Information Technology Implementation in Healthcare Facilities[45]	The HIT Maturity Model	HIT maturity	MoH
Jul 2018	Circular 39/2017/TT-BTTTT on Technical Standards for IT Implementation in State Organizations[46] (Replaced Circular 22/2013/TT-BTTTT)	The Recommended Standards for IT in State Organizations	Standards for IT applications in state organizations	Ministry of Information and Communications (MIC)
Mar 2018	Circular 48/2017/TT-BYT on Regulations on data exchange in management and reimbursement of health insurance claims[47]	The Electronic Claim Regulations	Electronic health insurance claim	MoH
Dec 2018	Decision 7603/QD-BYT year 2018 on The Service Coding System for Healthcare Management and Health Insurance Reimbursement version 6[48]	The Terminology and Service Coding System version 6	Electronic health insurance claim	MoH
Oct 2019	Decision 4888/QD-BYT year 2019 on The Smart Health IT Implementation and Development Scheme from 2019 to 2025[49]	The Smart HIT Scheme	Digital health strategies	MoH

Mar 2019	Circular 46/2018/TT-BYT on Regulations for Electronic Medical Records[50]	The Regulations for EMRs	Electronic medical records	MoH
Nov 2019	Decision 5349/QD-BYT year 2019 on Implementation plan for electronic health record[51]	The EHR Plan	Electronic health records	MoH
Dec 2019	Decision 6085/QD-BYT year 2019 on the eHealth Architectural Framework version 2.0[52]	The EHAF version 2	Architectural framework	MoH
Dec 2020	Decision 5316/QD-BYT year 2020 on The Digital Transformation in Health Care Scheme until 2025 and navigated towards 2030[53]	The Digital Transformation Scheme	Digital health strategies	MoH
May 2020	Decision 2153/QD-BYT year 2020 on Regulations on creation, utilization and management of health ID[54]	The Health ID Regulation	National health ID	MoH

[a]Applicable since March 2015 for organizations that had not implemented any health information systems, and since January 2017 for organizations that had implemented health information systems.

Policies: results of individual sources of evidence

Twenty government documents were combined into 11 groups based on the digital health domains or subjects that they addressed. These groups include hospital information management systems, general and interoperability standards, cybersecurity in health organizations, conditions for provision of health IT, electronic health insurance claims, laboratory information systems (LISs), HIT maturity, digital health strategies, electronic medical records, electronic health record, and eHealth architectural framework (EHAF). Table 4 presents these policy groups and the main purpose of each policy. In the following section, we summarize the policies and their objectives in chronological order.

Most of the digital health policies reviewed were released and came into effect after 2013, excluding the HIMS Guidance, which was enacted in 2006. The HIMS Guidance was an instruction regarding the functionalities and standards for HIMS used in state health facilities[35]. In 2013, the MoH promulgated guidance for interoperability standards and terminology systems applicable for health information systems[36]. Some of these are of required adoption, including the HL7 messaging version 2 or 3, DICOM, SDMX-HD, and ICD-10-CM, while the others are recommended, including the HL7 CDA, HL7 CCD, ATC, LOINC. Standards for general IT applications are guided by the Ministry of Information and Communications (MIC)[46].

The three cybersecurity guidance documents for state health organizations were published in 2014 and 2015. The Cybersecurity Guidance covers 16 components of an IT system in a health organization such as network, databases, applications, account management, and data transmission[37]. As organizations employing IT systems are required to develop and implement their own cybersecurity policy, the Guidance for Local Cybersecurity Policy provides a general instruction for the making of these local policies[39]. Lastly, the Guidance for Cybersecurity Response is a standard procedure to identify, classify, report and handle cybersecurity issues in health facilities[40]. In March 2015, the Required Conditions for HIT announced the essential criteria that organizations must meet when implementing HIT in Vietnam[38]. Accordingly, organizations must satisfy specific requirements for IT infrastructure, information security, IT workforce, and implementation of several technologies.

In 2016 and 2017, the MoH and the Vietnam Social Security (VSS) initiated the electronic social health insurance claim system, which was accompanied by the promulgation of several regulations and guidance. First, the Social Health Insurance EAF was published, describing the system's structure and operation[41]. An online portal was established, on which health providers can submit claims and check insurance information. To provide instructions for health facilities on using the portal, the VSS published the Insurance Portal V2 Guidance[42]. As claim data must be collected and formatted in a standardized manner, the Claim Standardization Guidance is intended to provide the necessary instructions for health facilities on claim preparation[43]. The Vietnam social health insurance system utilizes a national terminology and service coding system to form billing codes from medical documentation. In 2018, this coding system was in its 6th iteration[48]. Lastly, the Electronic Claim Regulations defines the responsibilities of health providers and insurance agencies surrounding claim sharing and feedback[47].

A guidance for laboratory information systems was published in August 2017, addressing the functionalities and standards required for such systems in Vietnam[44]. From 2018 to 2020, the MoH issued several policies that govern key digital health systems such as the maturity model for HIT, EMRs, and EHRs. From February 2018, the HIT Maturity Model began to be used as a roadmap for healthcare organizations to build and upgrade their digital health system[45]. Similar to the HIMSS Analytics Electronic Medical Record Adoption Model[55], this maturity model features seven levels of HIT application, each of which is built on a specific set of criteria. The most basic level (Level 1) only requires the implementation of HIS to manage outpatient services while the highest level (Level 7) defines a paperless environment that deploys a comprehensive range of HIT such as HIS, EMR, LIS, radiology information system & picture archiving and communication systems (RIS-PACS), and CDSS with interoperability between the systems. EMR adoption is governed by the Regulations for EMRs, enacted in March 2019[50]. The circular sets out the requirements for EMRs so that their use abides by Healthcare Law and is eligible to completely replace paper medical records. In the same year, the MoH announced a plan to roll out EHRs on a nationwide scale, starting in municipal cities[51]. While EMRs serve inpatient care and can vary in their specifications depending on the specialty in which they are implemented, the EHR system can be seen as the digital version of the paper Primary Health Record[56] and is a lifetime health document of every Vietnamese resident. Information in EHRs is mainly collected and updated by the district health facilities and outpatient clinics. The MoH has also announced the National Health ID system in the Health ID Regulation[54], in which each health ID is unique and represents an individual for their lifetime.

An overall picture of the IT systems that the MoH has been developing is depicted in their EHAF, presented in Figure 4. The framework was first published in 2015[57] and subsequently updated in 2018[58] and in 2019 with the EHAF Version 2[52]. According to the EHAF Version 2, EMRs and EHRs are the two main components of the healthcare building block. In 2019, the MoH announced the Smart HIT Scheme, presenting an agenda to develop and implement digital and smart technologies in Vietnam's health care for the period from 2019 to 2025[49]. By the end of 2020, the Digital Transformation Scheme was published, setting the goals to implement IT in multiple areas of the health sector including state administration, cashless payment and telehealth, disease prevention and primary care, and healthcare[53].

Table 4. Main purposes of the policies

Group of domains	**Denoted title**	**Purpose of the document**	**Valid from**

Hospital information management system	The HIMS Guidance	A guidance for functionalities and standards of hospital information management systems	Dec-06
General and interoperability standards	The Recommended Standards for HIT	A list of nomenclature systems and interoperability standards that the Ministry of Health (MoH) required or recommended for health information systems	Jun-13
	The Recommended Standards for IT in State Organizations	A list of general IT standards that the Ministry of Information and Communications (MIC) required or recommended for health information systems	Jul-18
Cybersecurity in health organizations	The Cybersecurity Guidance	A comprehensive guidance of cybersecurity measures for health facilities	Oct-14
	The Guidance for Local Cybersecurity Policy	A guideline for developing organization information safety and security policies for health facilities	Oct-15
	The Guidance for Cybersecurity Response	A guidance for classifying, identifying, reporting and handling cybersecurity issues in health facilities	Oct-15
Conditions for provision of health IT	The Required Conditions for HIT	Criteria that health facilities must satisfy when implementing digital health systems. Four areas addressed in the circular include: IT infrastructure, information security, human resource, and specific criteria for some health information technology (HIT) systems	Mar 2015 or Jan 2017
Electronic health insurance claim	The Social Health Insurance EAF	Explaining the architecture model for the social health insurance information system to be built by the MoH	Sep-16
	The Insurance Portal V2 Guidance	Announcing the launching of the Health Insurance Portal version 2 with an installation manual attached	Jun-16
	The Claim Standardization Guidance	A guidance for standardization of claims data including variable definition and data standards.	Sep-17
	The Electronic Claim Regulations	Responsibilities of the healthcare organizations and the insurance agencies in electronic claim exchange and investigation.	Mar-18
	The Terminology and Service Coding System version 6	A common list of health services with the relevant codes used in the social health insurance claim and reimbursement, updated to version 6.	Dec-18

Laboratory Information System	The LIS Guidance	A guidance for functionalities and standards of laboratory information systems	Aug-17
HIT maturity	The HIT Maturity Model	Seven levels of HIT application applicable for healthcare organizations, made of eight key components and capabilities. Criteria for each HIT levels and components were provided herein.	Feb-18
Digital health strategies	The Smart HIT Scheme	Presenting the agenda to develop and implement digital and smart technologies in Vietnam's health care for the period from 2019 to 2025	Oct-19
	The Digital Transformation Scheme	Presenting the agenda to comprehensively implement IT in the Vietnam's health care until 2025 and navigated towards 2030	Dec-20
Electronic Medical Records	The Regulations for EMRs	Criteria for EMR development and implementation to abide by healthcare law and replace paper medical records	Mar-19
Electronic Health Record	The EHR Plan	The national plan of the MoH to build and implement the EHR system in Vietnam	Nov-19
	The Health ID Regulation	The health ID system used for electronic health data of Vietnamese residents	May-20
Architectural framework	The EHAF version 2	Explaining the architecture model of key IT systems and databases built by the MoH	Dec-19

Policies: details of included policies

In the section below, we expand on the key policies in each digital health domain.

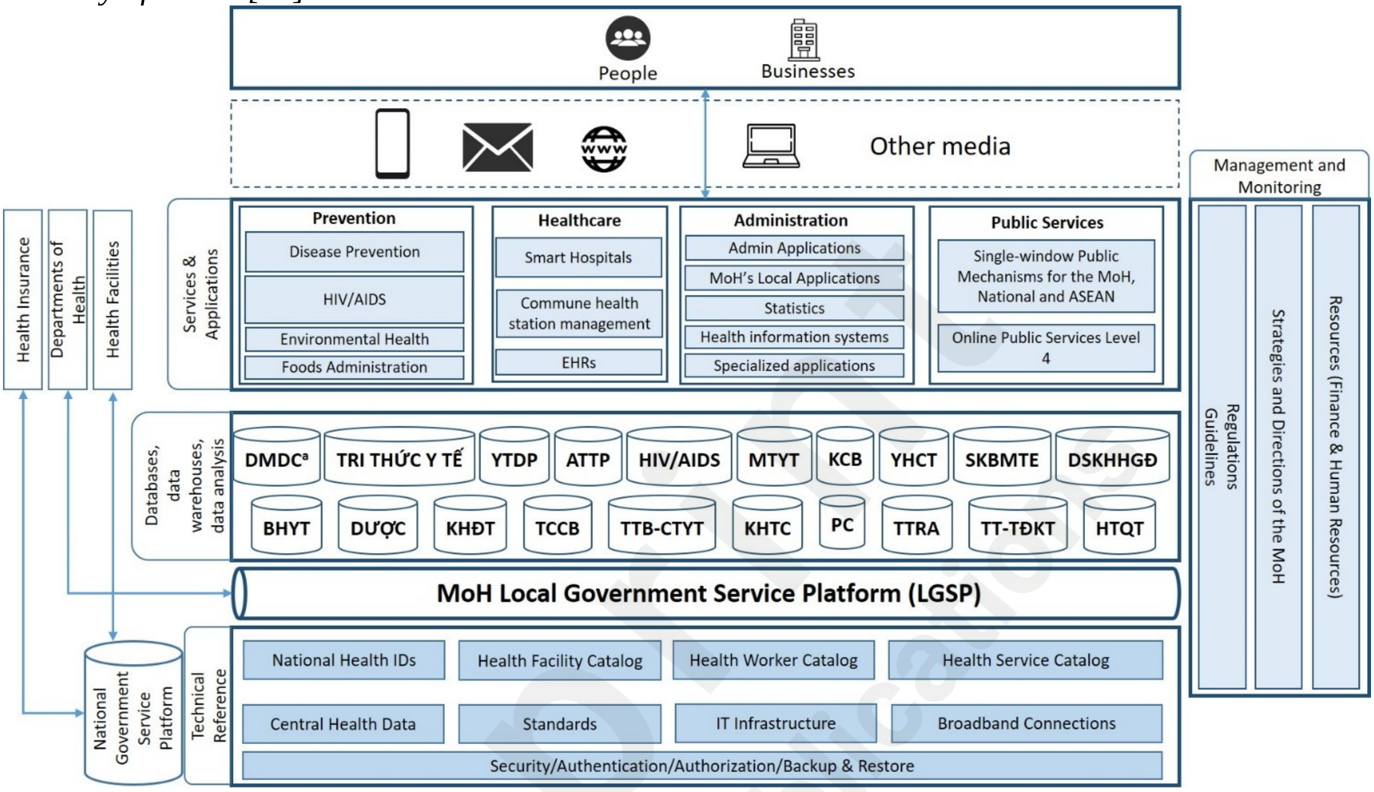
1. Architectural Framework: The EHAF version 2 (Decision 6085/QD-BYT year 2019 on the eHealth Architectural Framework version 2.0[52])

The framework is intended to guide the future developments of eHealth in the public sector, ensuring their compatibility with the current components and directed towards the common goals.
The EHAF version 2.0 has seven layers including:

- End-users (people and businesses).
- Communication channels (means to communicate with applications and services of the MoH such as computers, smart phones, and information portals).
- Services and applications layer. This layer comprises four building blocks including Disease Prevention, Healthcare, Administration, and Public Services.
- Databases and data analysis tools layer. This includes multiple databases specialized for each health area managed by the MoH, e.g., traditional medicine, pharmacy, and health insurance.
- The MoH's Local Government Service Platform (LGSP). This platform provides the shared supporting services for the upper layers. The MoH's LGSP is also able to communicate with other ministries and provinces through the National Government Service Platform.
- Technological infrastructure layer.
- Management and monitoring layer.

An illustration of the EHAF version 2.0 can be found in Figure 4.

Figure 4. The MoH's eHealth Architecture Framework version 2.0. Adapted from Vietnam Ministry of Health[52]



[a]**English translation of the abbreviations**: DMDC (reference catalogs), Tri thức y tế (Health knowledge), YTDP (Preventive medicine), ATTP (Food safety), HIV/AIDS (HIV/AIDS), MTYT (Environmental health), KCB (Healthcare), YHCT (Traditional medicine), SKBMTE (Maternal and child health), DSKHHGD (Population planning), BHYT (Health Insurance), DƯỢC (Pharmacy), KHĐT (Research and training), TCCB (Human resources), TTB-CTYT (Medical devices and infrastructure), KHTC (Finance), PC (Legislation), TTRA (Inspection), TT-TĐKT (Reward), HTQT (International collaboration.

2. Digital Health Strategies

2.1. The Smart HIT Scheme (Decision 4888/QD-BYT year 2019 on The Smart Health IT Implementation and Development Scheme from 2019 to 2025[49])

The Smart HIT scheme seeks to employ digital health, particularly smart technology, in Vietnamese health care. The scheme highlighted the Industrial Revolution 4.0 concept and the technologies characterizing this era, e.g., big data, artificial intelligence, and the internet of things. Following the intention of the Vietnamese government to take advantage of this revolution, the scheme points out the readiness of Vietnam's health system for implementing digital health technologies along with the impact that these technologies can have on the health system. Lastly, the scheme presents an agenda to promote and implement digital and smart health IT in Vietnam. The information relevant to the scope of this study is summarized below.

General goals

The scheme aims to "implement and develop digital health and smart health technologies for a modern, high-quality, equitable, efficient, and internationally integrated health system" and "to promote residents' access to health information so that they can use a highly-efficient health service and have their health continuously protected, taken care of, and promoted during their lifetime".

Specific goals

1. Developing a smart health care and disease prevention system.
2. Promoting IT implementation in health facilities for administrative process improvement and reducing hospital overload: adopting EMRs to replace paper records, using cashless payment for hospital billings, and establishing smart hospitals.
3. Promoting IT implementation in health administration: installing the electronic office system, public portals, and single-window information system of administrative procedures, promoting level 3 and level 4 online public service, building a smart health administration.

Areas of action

The scheme proposes 9 areas of action from 2019 to 2025, including:

1. Building the regulatory framework, guidance, standards, and economic-technical norms.
2. Building the health IT infrastructure.
3. Building a smart health care and disease prevention system.
4. Building a smart healthcare system.
5. Building a smart health administration system.
6. Developing the workforce.
7. Promoting smart health IT research, development, and implementation.
8. International cooperation.
9. Educating the public's awareness of smart health care.

In the scope of this study, we presented the detailed agenda of Area 1, building the regulatory framework, guidance, standards, and economic-technical norms, and Area 4, building a smart healthcare system as follows.

Area 1 - Building the regulatory framework, guidance, standards, and economic-technical norms

- Building eHealth architecture framework as a prerequisite for IT implementation in the health system.
- Developing regulations for resident health ID.
- Developing standards for interoperability between HIT systems, i.e., health station management software, EMRs, and EHRs.
- Developing economic-technical norms for HIT, in which HIT costs are a part of the total health service cost.
- Building policy for managing and using electronic nomenclature and coding systems in the health system.
- Building policy for using EHRs.
- Developing regulations for cybersecurity and privacy protection for health information in the online environment.
- Building human resource policies for HIT specialists.

Area 4 - Building a smart healthcare system

- Updating the management software and digital health systems in hospitals.
  - Develop hospital information systems that adhere to the national and international standards, in which interoperability between the information systems, and between the information system and the digital devices (lab devices, imaging diagnosis devices, interactive screens, personal mobile devices, etc.) is ensured.
  - Standardizing the national health ID system.
  - Building smart hospitals – Healthcare organizations consult the HIT Maturity Model to develop their smart hospital roadmap (Level 6 in this circular).
- EMRs are implemented in all healthcare organizations according to the EMR roll-out timeline in Circular 46/2018/TT-BYT on the Regulations for EMRs, aiming for paperless medical records and cashless payments in hospitals.
- Establishing and scaling up information kiosks in hospitals.

- Promoting AI application in healthcare with the following priorities:
    - Building interoperability standards to implement Internet of Medical Things as an infrastructure to operate CDS systems.
    - Developing real-time CDS systems closely integrated with EMRs.
    - Imaging diagnosis assistance.
    - Surgery assistance.
    - Encouraging businesses and healthcare organizations to build big data systems embedded with AI algorithm to support clinical decision making.
    - Disease diagnosis, treatment, and prevention with traditional medicines.
    - Applying AI in specialties such as imaging diagnosis, cardiovascular diseases, respiratory diseases, orthopedics, cancer, obstetrics, and paediatrics.

2.2. The Digital Transformation Scheme (Decision 5316/QD-BYT year 2020 - The Digital Transformation in Health Care Scheme until 2025 and navigated towards 2030[53])

The Digital Transformation Scheme seeks to implement IT in all aspects of health care as it defines digital transformation in health care as “the comprehensive application of information technology which prioritizes the cutting-edge digital technology that can make positive changes in all aspects of health care”.

Targets

The key four areas addressed in this scheme are state administration, cashless payment and telehealth, disease prevention and primary care, and healthcare. In the healthcare area, the scheme aims for 15% (210 hospitals) and 50% (700 hospitals) of hospitals in the country to successfully adopt paperless EMRs and cashless payment by 2025 and 2030 respectively.

Areas of actions

Table 5 summarizes the areas and sub-areas of action set out in the scheme.

Table 5. Summary of areas of action in Decision 5316/QD-BYT year 2020

1. Infrastructure development	1.1. Awareness education
	1.2. Building the regulatory framework, guidance, standards, and economic-technical norms
	1.3. Building and upgrading the IT infrastructure
	1.4. Building health databases
	1.5. Building digital health platforms
	1.6. Ensuring cybersecurity
	1.7. International cooperation, research, and innovations in digital health
	1.8. Workforce development
2. Implementing IT in administration and public services	
3. Promoting investments in digital health from businesses and hospitals	
4. Digitalization in societies	
5. Digital transformation in primary care and disease prevention	
6. Digital transformation in hospitals	

Area 5 (digital transformation in primary care and disease prevention) and Area 6 (digital transformation in hospitals) are considered the priorities over the others. We present details of area 1.2 and area 6 as follows.

Area 1.2 - Building the regulatory framework, guidance, standards, and economic-technical norms

- Publishing regulations and guidance for doing trials of novel digital health products; developing digital health platforms.

- Developing regulations and standards for data exchange and interoperability between health information systems based on international standards.
- Developing regulations for collecting and managing health data; building a decree for the national health database.
- Providing guidance for digital health technologies; updating guidance for building smart hospitals and paperless hospitals.
- Developing regulations for protecting security, safety, and confidentiality of health data in the online environment.
- Developing regulations and guidance for electronic authentication in health care.
- Implementing and updating the MoH's e-government structure.
- Developing financial mechanisms for health IT services as part of the overall healthcare service and mechanisms for hiring health IT services.
- Developing policies and regulations for telemedicine and electronic prescription so that patients can utilize remote healthcare services.

Area 6 - Digital transformation in hospitals

This area is largely similar to Area 4 (Building a smart healthcare system) of the Smart HIT Scheme. A comparison between these is shown in Table 5.

Table 6. Comparing Area 4-Decision 4888/QD-BYT year 2019 and Area 6-Decision 5316/QD-BYT year 2020

	Area 4 of the Smart HIT Scheme	**Area 6 of the Digital Transformation Scheme**
Updating management software and digital health systems in hospitals	Yes[a]	Yes
Standardizing the national health ID system	Yes	Yes
Building smart hospitals based on Circular 54/2017/TT-BYT	Yes	Yes
EMR implementation based on Circular 46/2018/ TT-BYT	Yes	Yes
Building information kiosks	Yes	No
Promoting AI application in healthcare	Yes	Yes
Conducting telehealth and online registration based on Decision 2628/QD-BYT year 2020	No	Yes
Implementing the national prescription management system in all healthcare organizations	No	Yes

[a]"Yes" means the information was addressed in the document while "No" means not addressed.

3. Electronic Medical Records: *The Regulations for EMRs (Circular 46/2018/TT-BYT on Regulations for electronic medical records[50])*

EMR refers to the clinical record system used in hospitals' inpatient departments. They are locally managed by healthcare institutions and different from the EHR system, which is centrally managed on the national database and intended for use by the commune health facilities and outpatient clinics.

This circular is the first official guidance for EMRs in public healthcare facilities. It ensures that the use of EMRs can uphold the equivalent law and can legitimately replace paper medical record. Healthcare organizations are eligible to discontinue paper medical record keeping if they can satisfy all the criteria in this circular, which concern the following areas:

1. Content specifications
2. EMR creation and update
3. Storage and backup
4. Access right and secondary use
5. Patient identification
6. Digital/electronic signatures
7. Basic functionalities
8. Adherence to interoperability and IT standards
9. Data security and confidentiality
10. Nomenclature system

Three essential criteria that must be prioritized are:

- Each patient's EMR is provided with an ID number that is unique in the health organization.
- All information that a traditional medical record collects can also be recorded by EMRs.
- Each EMR must have a digital signature of the one who is responsible for the information entered in that record.
- Data stored in an EMR system is protected by Section 2, Article II of Cyberinformation Security Law[59].

In addition, the circular also provides the criteria for paperless lab test and imaging diagnosis management with LIS and picture archiving and communication system (PACS). Full details of this document can be found in Multimedia Appendix 2.

4. Health Information Technology Maturity: The HIT Maturity Model (Circular 54/2017/TT-BYT on Assessment Criteria for Information Technology Implementation in Healthcare Facilities[45])

The MoH put into effect the HIT Maturity Model as a roadmap for HIT implementation in healthcare facilities. Healthcare organizations are required to evaluate their current HIT maturity using this model and report the evaluation results to the MoH. The baseline results are a requisite for healthcare organizations to set goals for their next maturity level. An overall HIT maturity level depends on the maturity level of the individual domains, which are:

1. IT Infrastructure
2. Administration and Operation Software
3. Hospital Information System (HIS)
4. Radiology Information System – Picture Archiving and Communication System (RIS-PACS)
5. Laboratory Information System
6. Non-functional Standards
7. Security and Information Safety
8. Electronic Medical Record

In addition, there are specific capabilities that are required for each maturity level. An organization's overall HIT maturity can range from "HIT Level 1" to "HIT Level 7".

There are seven levels (1 to 7) applicable for the IT infrastructure domain and the HIS domain, and two levels (basic and advanced) applicable for the other domains (Administration and Operation Software, LIS, Non-functionality Standards, Security and Information Safety, RIS-PACS, and EMRs).

Below are summaries of the seven levels of HIT maturity. An overview of all the seven levels can be found in Figure 5.

HIT Level 1

HIT Level 1 includes Level 1 of IT Infrastructure and Level 1 of HIS, which aims to provide electronic access to patient data. This IT infrastructure level essentially requires workstation computers, a local area network and Internet connection while HIS manages outpatient and pharmacy data, and sends claim data to the health insurance claim system. Adoption of MoH's terminology and service coding system is compulsory.

HIT Level 2

HIT Level 2 is an upgrade of IT Infrastructure and HIS from Level 1 to Level 2. This will add to the pre-existing system a dedicated server and the laboratory modules that manage lab orders and results. HIT Level 2 particularly aims at building a clinical data repository (CDR) of pharmacy data, lab orders, test results, and the terminology and service coding system. Data in the CDR can be accessed by multiple stakeholders engaging in patient care.

HIT Level 3

HIT Level 3 requires an upgrade of IT Infrastructure and HIS from Level 2 to Level 3. Added components include LIS, administrative applications, and cybersecurity solutions at a "Basic" level. At this level, more security and storage infrastructures are provided, including firewall devices, security solutions and specialized storages. CDR capacity is increased to cover vital sign data, nurse notes, and medical procedures. Healthcare organizations would start implementing CDS system Level 1 to assist doctors with drug information via the facility's intranet.

HIT Level 4

HIT Level 4 comprises Level 4 IT infrastructure, Level 4 HIS, Advanced LIS, and Basic level of RIS-PACS, administration and operations software, cybersecurity, and non-functional standards. Organizations at this HIT level can manage their imaging diagnosis data electronically using RIS-PACS. Although a Basic RIS-PACS has yet to discontinue physical film archiving[50], it offers essential functionalities such as integration with imaging diagnosis machines and HIS, digital image viewing, and measurement. An Advanced LIS can manage test samples and supplies, integrate with the HIS, and send out alerts for abnormal test values. A storage network (SAN or NAS), a queue management system with display screens will complement the IT infrastructure. Outpatient orders from doctors can be made electronically while all inpatient orders will be managed in the digital system.

HIT Level 5

HIT Level 5 differs from HIT Level 4 in Level-5 IT infrastructure, Level-5 HIS, and "Advanced RIS-PACS". The Advanced level of RIS-PACS is able to conduct multi-site consultation, allowing images to be viewed on multiple devices such as laptops and mobile phones. Furthermore, the Regulations for EMRs allows paperless image management if healthcare facilities have "Advanced RIS-PACS" with an eligible storage capacity[50]. At Level 5, HIS can manage emergency rooms, operating theatres, appointments, follow-ups, and can integrate electronic patient card.

HIT Level 6

Transitioning from HIT Level 5 to HIT Level 6 requires organizations to implement a "Basic" EMR system in addition to upgrading their IT infrastructure, HIS, Administration and Operation software, Security and Data Safety, CDSS, drug management, and non-functional standards to an "Advanced" level. Hospitals qualified for HIT Level 6 or higher are certified as "Smart Hospital". The Basic EMR system provides inpatient medical records and can integrate with other clinical information systems. Mobile devices working on wireless LAN, hospital security cameras, and backup storage systems are needed to meet Level 6 IT infrastructure.

Hospitals at HIT Level 6 can receive and utilize data from multiple information systems such as drug information and interactions, treatment protocols, and nutrition. Clinical information can be accessed via mobile devices. Additionally, HIT Level 6 features several extra capabilities including Level 2 CDSS, a safe medicine management procedure, and increased digitalization of clinical documents. Level 2 CDSS can warn doctors of drug interactions and potential treatment risks based on the most updated evidence. The medicine management system provides a closed and automatic medicine management procedure using automated dispensing systems and digital identification devices, e.g., barcodes and Radio Frequency Identification (RFID). All structured clinical forms such as progress reports, consultation notes, problem lists, and discharge summaries are digitized.

HIT Level 7

At HIT Level 7, all the eight domains are at their highest level. Organizations are now equipped with information kiosks and network monitoring systems. A Level 7 HIS provides functionalities to manage EMRs, professional procedures, control information kiosks, and support cashless payment. EMR systems at “Advanced” level have dedicated functionalities to manage medical records including increased storage duration, syncing records, and restoring records. Data stored in the EMR system is better protected and can be exchanged under international standards such as HL7. According to the Regulations for EMRs[50], an “Advanced” EMR system with qualified storage capacity is eligible to replace the paper-based medical record system. Data from CDR are analysed to improve quality of care, patient safety and service efficiency. Insights for operational activities can be continuously generated to inform hospital departments such as inpatient, outpatient, and emergency departments. Organizations at this HIT Level are able to exchange data using HL7 standards. Lastly, Level 3 CDSS is able to support a wider range of diagnosis and treatment decisions.

Figure 5. Levels of HIT Maturity

	IT infrastructure	HIS	LIS	RIS-PACS	EMR	Administration and Operation Software	Security and Information Safety	Non-functional Criteria	Extra capabilities
HIT Level 7	Level 7	Level 7	Advanced	Advanced	Advanced	Advanced	Advanced	Advanced	- "Paperless" hospital if all relevant criteria are met - CDSS level 3 supporting doctors' decisions related to treatment protocols and treatment results using suitably customized templates - Data in CDR is analysed to improve care quality, patient safety, and care efficiency - Clinical data can be readily shared for stakeholders in care coo rdination based on HL7 standards - Continuous reports of hospital services using the data collected
HIT Level 6 (Smart hospital)	Level 6	Level 6	Advanced	Advanced	Basic	Advanced	Advanced	Advanced	- CDSS level 2 providing: evidence-based warnings for treatment i.e. health and medicine advices, drug information/interaction check, and initial order/prescription violation indentification rules - All structured forms i.e. progress notes, consultation notes, problem lists, and discharge summaries are digitalized - Closed-loop management of drugs, using identification technologies to assist drug administration
HIT Level 5	Level 5	Level 5	Advanced	Advanced	n/a	Basic	Basic	Basic	PACS can replace physical films
HIT Level 4	Level 4	Level 4	Advanced	Basic	n/a	Basic	Basic	Basic	- PACS allows doctors to access images outside the imaging department - Electronic ordering - Electronic management of inpatient orders
HIT Level 3	Level 3	Level 3	Basic	n/a	n/a	Basic	Basic	Basic	- Electronic records having digital vital sign records, nursing notes, medical procedures, and surgical procedures are stored in CDR - CDSS level 1 assisting electronic prescription (new or historic prescription) - Pharmacy information available in the hospital network and supported with CDSS
HIT Level 2	Level 2	Level 2	n/a	n/a	n/a	n/a	n/a	n/a	- A CDR consisting the nornenclature and coding systems, pharmacy, orders, and test results (if available) - Data in CDR can be shared between stakeholders for care coordination
HIT Level 1	Level 1	Level 1	n/a	n/a	n/a	n/a	n/a	n/a	Patient's information can be accessed electronically

[a]HIT: health information technology; HIS: hospital information system; LIS: laboratory information system; RIS: radiology information system; PACS: picture archiving and communication systems; EMR: electronic medical record; CDSS: clinical decision support system; CDR: clinical data repository.

5. Conditions for Provision of Health IT: The Required Conditions for HIT (Circular 53/2014/TT-BYT on Required conditions for provision of health IT activities[38])

The Required Conditions for HIT can be seen as a summary of the criteria that healthcare facilities need to satisfy when implementing electronic health systems.

This circular announces the conditions and requirements that healthcare facilities working in Vietnam must abide by when implementing and operating health IT systems. Health IT activities, as defined in the circular, involve “providing, transferring, collecting, analysing, storing and exchanging health care data through the IT infrastructure”. The conditions specify four areas related to health IT implementation: IT infrastructure, information security, human resource, and operational requirements (see Multimedia Appendix 3 for details).

6. General and Interoperability Standards

This group included two guidance documents, one from the MIC and one from the MoH. While the former addressed the general technical standards that IT applications in state organizations need to adopt, the latter revolved around the national and international standards targeted at health information systems.

6.1. The Recommended Standards for HIT (Decision 2035/QD-BYT year 2013 – Terminology systems and data exchange standards recommended for health IT[36])

In June 2013, the MoH published Decision 2035/QD-BYT recognizing the terminology systems and technical standards applicable for health IT systems. Some terminologies and standards are required while others are recommended for adoption in public healthcare organizations. It is strongly advised in the decision that all public facilities should adopt these systems, and information systems not matching them should plan for relevant transitions. Table 6 presents the standards and nomenclature systems recommended in the decision.

Table 7. Standards and nomenclature systems for health IT systems

Categories	**Standard names**	**Reference**	**Guidance**
Administrative nomenclature	The list of official administrative units in Vietnam	Decision 124/2004/QD-TTg year 2004 and its amendments	Compulsory
	The list of ethnic groups in Vietnam	Decision 121-TCTK/PPCD year 2004 and its amendments	Compulsory
	The list of occupations in Vietnam	Decision 114/1998/QD-TCTK year 1998 and its amendments	Compulsory
International classification and coding for diseases and medical services	ICD-10-CM	WHO	Compulsory
	ICD-O-3	WHO	Recommended
	ICD-10-PCS	WHO	Recommended
	ATC	WHO	Recommended
	LOINC	Regenstrief Institute	Recommended
International interoperability standards	HL7 messaging version 2.x or 3.0	hl7.org	Compulsory
	DICOM version 2.0	medical.nema.org	Compulsory
	SDMX-HD	sdmx-hd.org	Compulsory
	HL7 CDA	hl7.org	Recommended
	HL7 CCD	hl7.org	Recommended
	ELINCS	elincs.chcf.org	Recommended

In addition to the standards above, the MoH instructs IT systems in the health sector to follow other relevant IT standards from the MIC that were addressed in Circular 01/2011/TT-BTTTT, which was then replaced by the new version, Circular 22/2013/TT-BTTTT.

6.2. The Recommended Standards for IT in State Organizations (Circular 39/2017/TT-BTTTT on Technical Standards for IT Implementation in State Organizations[46])

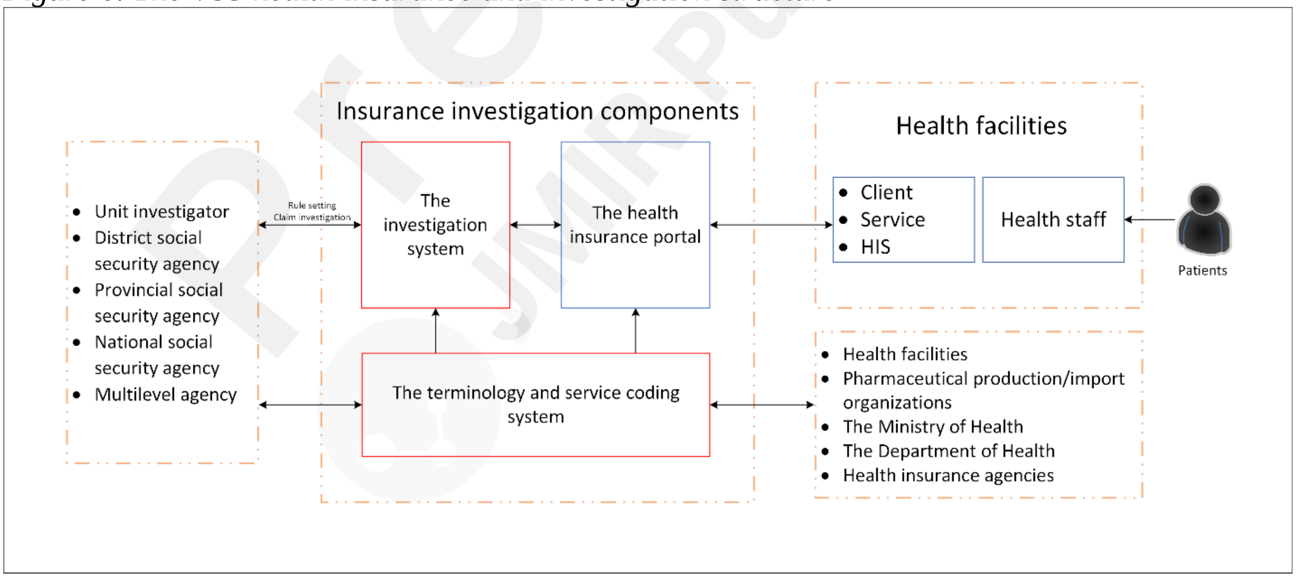
The MIC released Circular 39/2017/TT-BTTTT as a guidance for adoption of technical IT standards in state organizations. Its first version is Circular 22/2013/TT-BTTTT, released in 2013. The circular advises on five categories of standards such as connection, data integration, information access, information security, and public key infrastructure. The MIC encourages systems currently not adopting the standards in this circular to carry out transitions by July 2018. A detailed list of the standards is shown in the appendix. (Please see Multimedia Appendix 4 for full details)

7. Electronic Health Insurance Claim

7.1. The Insurance Portal V2 Guidance (Decision 917/QD-BHXH year 2016 - Announcement of the health insurance portal version 2[42])

Decision 917/QD-BHXH year 2016 from the VSS announces the health insurance portal version 2 and provides detailed guideline on connection configuration and structuring data sets. The guideline is intended for use by healthcare facilities and provincial health insurance agencies who are responsible for regular data upload to the portal. The VSS health insurance claim and investigation structure are presented in Figure 6.

Figure 6. The VSS health insurance and investigation structure



The guideline demonstrates three protocols that claim data can be submitted, each of which is accompanied by detailed instructions such as setting up connections, properly formatting data, and uploading data to the portal.

- Via web service connection - supports a variety types of data submission including clinical data, service validation, monthly reports, health insurance ID checking, examination history, and referral receipt from other facilities.
- Data sync from client software - supports syncing clinical data to the system.

- Direct data entry on the portal - supports clinical data entry to the system.

The portal also offers functionalities for checking health insurance account information, viewing information of previous hospital visits, and checking referral documents between healthcare facilities.

7.2. The Electronic Claim Regulations (Circular 48/2017/TT-BYT on Regulations on data exchange in management and reimbursement of health insurance claims[47])

The MoH mandated Circular 48/2017/TT-BYT as an official regulatory document for online health insurance claim and investigation for healthcare organizations. The circular essentially articulates the responsibilities of healthcare facilities and the health insurance agency as well as relevant rules in sending data and follow-ups. The fundamental technical requirements previously described in the VSS's guidelines are also outlined in the document. (Please see Multimedia Appendix 5 for further details).

7.3. The Claim Standardization Guidance (Decision 4210/QD-BYT year 2017 on Requirements for standard and format of output data used in management, investigation and reimbursement of insurance-paid health care expenses[43])

The health insurance portal requires claim data to be formatted as XML, encoded with UTF-8, and relevant data sets to be standardized for efficient management and assessment. To support organizations in preparing claim data, the MoH has delivered data formatting guides and continuously updated them. The latest release is found in Decision 4210/QD-BYT year 2017. Claims for groups of health services are recorded into separate data sets. Data sets from the same patient are linked through a unique ID number coded as *MA_LK*. The instructions include data item, data type, the allowed maximum length, and further guidance for the data item.

Table 8. Overview of guidance for data standardization

Tags	**Data type**	**Maximum length**	**Guidance/explanations**
Each data item is presented as a coded variable	Numeric, character, or other types	The maximum length for one row	The exact data to look for and/or how to collect or compile the data

The scope of the document is as follows:

- Guidance for claims of general health services
- Guidance for claims of medications
- Guidance for claims of medical procedures and supplies
- Guidance for claims of imaging diagnosis and lab test services
- Guidance for claims of follow-up interventions
- The official groups of services by cost
- The official hospital department codes
- The official accident and injury codes
- The official medicine bid packages and package codes.

7.4. The Terminology and Service Coding System version 6 (Decision 7603/QD-BYT year 2018 on The Service Coding System for Healthcare Management and Health Insurance Reimbursement version 6[48])

The Service Coding System for Healthcare Management and Health Insurance Reimbursement, or the terminology and service coding system, was first piloted in three cities in Vietnam in 2015[60]. Since then, it has been continuously updated, with version 6[48] being the most up-to-date. The system features nomenclatures of health services, their standardized codes, regulated prices, and the ICD-10 codes with Vietnamese instruction. The terminology and service coding system has been made compulsory for use in health information systems of state health facilities and online health insurance claim activities. The 11 nomenclature groups provided in Version 6 are:

- Technical service codes.
- Department codes.
- Inpatient bed codes.
- Half-day bed codes in chemotherapy and radiotherapy.
- Medication codes.
- Traditional medication codes.
- Traditional diagnosis codes.
- Medical supply codes.
- Blood product codes.
- ICD-10 codes.
- Lab test codes.

8. Cybersecurity in Health Organizations: The Cybersecurity Guidance (Decision 4159/QD-BYT year 2014 - Guidance on ensuring security of electronic health data in health organizations[37])

In October 2014, the MoH promulgated Decision 4159/QD-BYT year 2014 detailing the required measures to safeguard cybersecurity in health facilities. These regulations are applicable for organizations that conduct health IT activities in the online environment, including managing, using, storing, and exchanging health information. In general, the measures and recommendations addressed by this decision aim to maintain three characteristics in digital health implementation:

- Security:
    - Only authorized users can access to the health information.
    - Passwords and access keys are encrypted during access, transmission, and are saved at the healthcare facilities.
- Integrity:
    - Information can only be deleted or edited by authorized users. Information is preserved during storing and transmission.
    - Integrity is maintained in management, use, storage and transmission of information, in which changes are not allowed without authorization from the administration unit.
    - Measures to ensure integrity must be applied in accessing, entering, storing, using, processing, transferring, extracting, and recovering data.
- Availability:
    - Uninterrupted operation of the IT system is ensured.
    - Information can be accessed quickly in response to authorized requests.

- Human resource operating the IT system is ensured.
- Policies for managing and implementing the IT system are developed, publicized and adhered to.

(For further details, please see Multimedia Appendix 6)

9. Laboratory Information Systems: The LIS Guidance (Decision 3725/QD-BYT year 2017 - Guideline for implementation of laboratory information system in healthcare facilities[44])

Decision 3725/QD-BYT year 2017 provides guidance for the technical features and functionalities of a LIS used in healthcare facilities.

The guideline first highlights the essential points that organizations need to consider when implementing or developing a LIS:

- LIS's capability to communicate and integrate with other information systems in the hospital or healthcare system.
- Lab test results and related information can be exchanged and linked between different labs in a facility and with other facilities.
- LIS's design and technical documents to allow for easy repair, maintenance and upgrade.
- Technical standards recognized in Circular 22/2013/BTTTT and Decision 2035/QD-BYT year 2013, where applicable, must be adopted for LIS.
- Advanced technologies are encouraged for implementing LIS.
- Facilities must ensure the necessary conditions for managing and operating LIS such as safety, data security, IT infrastructure, and human resources.

(For further details, please see Multimedia Appendix 7)

10. Hospital Information Management System: The HIMS Guidance (Decision 5573/QD-BYT year 2006 – Requirements and functional modules for hospital management software[35])

In 2006, the MoH released the Decision 5573/QD-BYT year 2006 to guide the adoption of hospital management software (HMS) in Vietnam's public and private healthcare facilities. The guideline addresses the key requirements that an HMS should meet in the regulatory and clinical context of Vietnam. An HMS's functional modules and detailed instructions for each module are also presented in the document.

(For further details, please see Multimedia Appendix 8)

11. Electronic Health Records

The EHR system aimed to create and maintain a lifetime EHR for every Vietnamese resident. It differed from EMR systems which are managed by hospitals. EHR contents were developed based on the paper personal health record's standardized template. Each EHR profile is identified by a unique health ID.

11.1 The EHR Plan (Decision 5349/QD-BYT year 2019 – The Electronic Health Record Scheme[51])

In Decision 5349/QD-BYT year 2019, the MoH presents the scheme for setting up and implementing the

EHR system on a national scale. Accordingly, an EHR is the digital version of health records that are created, displayed, updated, stored, and exchanged using electronic devices. Health records are the medical documents that keep health care information of a person for their lifetime and are regulated by the MoH. The ultimate goal of this scheme is to provide every Vietnamese citizen with an EHR while step-by-step establishing a population health database in the National Health Data Center. The objectives and requirements set out in this scheme is shown below.

Requirements for EHR softwares

Table 9. Summary of requirements for EHR systems

The electronic health record (EHR) software's design must meet the required standards	The EHR software can collect all the data in a standard paper health record guided in Decision 831/QD-BYT year 2017 (The standard template of personal health record for primary health care)
	The EHR software can export data to XML files that follow the guidance set out in Decision 4210/QD-BYT year 2017 (Standards for claim data submitted to the online health insurance system)
	The EHR software can satisfy the regulations in Circular 48/2017/TT-BYT year 2017 (Regulations for data sharing in health insurance claim)
	Compatible with HL7 standards
	The EHR software can utilize the national health ID system and is interoperable with related health information systems
EHR functional modules	Health service provision functionality
	Administrative management functionality
	Information infrastructure management functionality
Personal information protection	Protection of personal data in EHRs abides by section 2, chapter II of the Electronic Data Safety Law

The unique patient identifiers

Every EHR profile will have a national health ID number that is unique for every citizen. The national health ID system is guided in Decision 2153/QD-BYT year 2020[54].

Data ownership and data management

Data generated in the EHR system is owned by government's administrative bodies: the MoH and the relevant Provincial Department of Health.

EHR providers or EHR developers have responsibilities to hand over data, software source codes and other tools that allow the MoH or the Department of Health to continue the EHR service with another provider.

EHR implementation plan

The scheme aims to reach an EHR coverage of 80% in the population of central-government provinces by the end of 2020, and a coverage of 95% in Vietnam's population by the end of 2025. Following are the steps to install and implement EHRs:

- Developing EHR software.

- Installing EHR software in health facilities.
- Providing EHR training for health workers.
- Creating EHRs and collecting data to EHRs using the pre-existing data at health facilities or via interviews.
- Continuously updating data to EHRs at health facilities.
- Making use of data collected in EHRs.
- Maintaining the EHR system.

Building the technical documents and guidance system

Technical documents and guidance to operate and manage EHRs will be built to guide EHR use at health facilities, including:

- Interoperability specifications with the national health ID system and related health information systems.
- Policies for using, managing, operating, and ensuring data safety for the EHR system.
- Policies for creating, updating and making use of EHRs.
- Financial mechanisms for maintaining and operating the EHR system.

11.2. The Health ID Regulation (Decision 2153/QD-BYT year 2020 on Regulations on creation, utilization and management of health identification[54])

Decision 2153/QD-BYT year 2020, released in May 2020, announced that the social insurance ID will be the National Health ID. These IDs are nationally unique for each patient and represents a set of identification data for that patient. Each identification data set includes the patient's full name, date of birth, gender, place of birth ID, and health insurance ID. Health IDs are intended to be used with EHR, EMR, and other health information systems as a common patient identifier system.

Discussion

This review explored the current state of digital health research and policies in Vietnam to inform the implementation of digital systems used in hospital care. Nearly half of the hospital-based studies were case reports of engineering solutions; one study assessed physicians' performance with assistance from a CDS software; two explored readiness to adopt EHRs; and two provided a high-level review of Vietnamese eHealth. The data analyzed in these studies were mostly collected in 2016 or earlier. These findings suggest a lack of research studies that investigate and inform the implementation of digital health systems in the Vietnamese hospital settings.

Government policies in Vietnam have paid significant attention to digital health over the past five years. The MoH has set out specific regulations for EMR use and a maturity framework that shapes the development of digital health systems in hospitals. Other national projects have taken place such as the electronic insurance claim system, the nationwide roll-out of an EHR, and a national health ID system. International standards such as HL7v2 messaging, ICD-10, and, recently, HL7 FHIR, have been consistently recommended. Guidance documents have also addressed measures to ensure cybersecurity in health organizations. These policies are guided by an architectural framework and digital health strategies which ultimately aim to leverage IT, especially smart technology, in all aspects of the sector.

Of all the policies reviewed, the Regulations for EMRs and the HIT Maturity Model can be seen as the key policies for HIT application in Vietnam's hospitals. While the previous national efforts focused on administrative systems such as HIMS[35], these policies aimed to promote adoption of clinical systems such as EMR, LIMS, and RIS-PACS. At this nascent stage of HIT adoption, Vietnam has been focusing on building the necessary infrastructure and emphasizing the role of HIT in achieving a more efficient paperless environment. This is reflected in the recent digital health agenda, which highlighted the crucial role of EMRs in replacing the paper medical records[49,53]. A major theme that the Regulations for EMRs addressed is the eligibility of transitioning to a complete paperless environment for EMR systems, as well as LIS and RIS-PACS[50]. While this is a practical and well-defined motivation for EMR adoption, the World Health Organization recommended the true benefits of EMRs, such as improved data quality, timely access to information, and increased care quality, should be proven to encourage buy-in from healthcare workers[61]. Thus, future policies may seek to link EMR and HIT adoption with such benefits through clearly-defined quality indicators to ensure meaningful HIT utilization. Lessons learned from some countries also recommend that financial incentives, when possible, should be made to increase EMR adoption[62–64].

The “HIT Maturity Model” defined a wide spectrum of HIT implementation levels that are applicable for hospitals of various digitisation stages. While the “Smart HIT Scheme” and the “Digital Transformation Scheme” aimed to establish smart hospitals (HIT Level 6) in the next few years, there may be a need for further policy development that takes into account the Vietnamese context as an LMIC. The “Assessment of HIT Maturity” conducted in 2019 showed a large dispersion of HIT maturity across the top-level hospitals directly governed by the MoH, among which nearly 20% were only at HIT Level 1, most hospitals (nearly 50%) were at HIT Level 3, and only one in the 42 hospitals was at HIT Level 5[65]. A readiness assessment conducted by the MoH in 2017 revealed that among 36 MoH-governed hospitals, only 30.6% had EMR systems, 38.9% had some forms of CDS software, and over half of these hospitals lacked specialized IT workforce and information security capacity[66]. Given the low HIT maturity levels that these high-rank hospitals held, HIT implementation and readiness in smaller hospitals can be even poorer. It is likely that rural hospitals will struggle to reach even the most basic levels of digital maturity due to a lack of funding, IT workforce, and infrastructure. Therefore, future policies may need to recommend further support for small and rural hospitals to reduce these gaps in digital maturity.

The two digital health strategies aim to promote AI in healthcare through applications such as CDSS integrated with EHR systems, imaging diagnosis and surgical support. New regulations, guidance, and standards are to be released to inform these adoptions[53]. Some of these intended policies are regulations for trials of novel digital health products, guidance for digital health technologies, and guidance for protecting online health data, to name a few. New regulation frameworks, along with an increased use of EMRs and EHRs in Vietnam’s hospitals, will be important facilitators for development and implementation of AI for hospital care in Vietnam. However, there is currently a lack of research to inform appropriate and sustainable use of these technologies in Vietnam. Local context factors such as disease burden, healthcare practices, and infrastructure constraints should be taken into consideration for future AI research and implementation, as highlighted by Schwalbe[5], Wahl[6], and Alami[67]. Future digital health research should seek to investigate these factors to best guide HIT and AI development and application in Vietnam’s hospitals.

Limitations

This review was limited to the context of hospital-based digital health systems and does not cover the perhaps more extensive academic literature on community-based systems and the use of national systems for monitoring infectious disease outbreaks and other public health interventions. Academic studies were only searched on international databases hence this study did not cover publications in Vietnamese journals that are not registered in these databases.

At the time this study was conducted, the digital health policy landscape in Vietnam is experiencing significant changes with new policies likely to be released and the existing policies may be subject to modifications over the next year. Therefore, this scoping review should be viewed as a snapshot of this area at the end of 2020.

Conclusions

Quantity and areas of research studies about digital health systems in Vietnam's hospitals is limited with few evidence to inform the implementation of new technologies in hospital care. The policies developed over the past five years to inform the adoption of digital health systems in Vietnam are comprehensive and will be useful for hospitals. They focus on guiding the basic functionalities and largely follow international standards and guidelines. Further research is needed to ensure that policies and guidelines are able to deal in detail with issues not encountered in HIC settings or that may be specific to the Vietnamese context.

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Authors' contributions

	DMT	CLT	JIVN	JM	AL	CP
Substantial contributions to the conception OR design of the work; OR the acquisition, analysis, OR interpretation of data for the work	Yes Yes Yes Yes	Yes Yes Yes	 Yes Yes	 Yes Yes	 Yes Yes	Yes Yes Yes
Drafting the work or revising it critically for important intellectual content	Yes	Yes	Yes	Yes	Yes	Yes
Final approval of the version to be published	Yes	Yes	Yes	Yes	Yes	Yes
Agreement to be accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved	Yes	Yes	Yes	Yes	Yes	Yes
Funding acquisition		Yes		Yes		Yes

Conflicts of Interest

None declared.

Abbreviations

AI: artificial intelligence
CDR: Clinical data repository
CDS: clinical decision support

EHAF: eHealth architectural framework
EHR: electronic health record
EMR: electronic medical record
HIMS: hospital information management system
LGSP: local government service platform
LIS: laboratory information system
LMICs: low-and-middle income countries
MIC: Ministry of Information and Communication
MoH: Ministry of Health
PRISMA-ScR: Preferred Reporting Items for Systematic reviews and Meta-Analyses extension for Scoping Reviews
VSS: Vietnam Social Security

Multimedia Appendix 1

Search Strategy

Governments' policies and programmes were searched on thuvienphapluat.vn database using the following search terms:

1. "-BYT" "điện tử" (search date: March 5th 2020; search in title)

2. "-BYT" "bệnh án" (search date: March 5th 2020; search in title)

3. "-BYT" "công nghệ" (search date: March 5th 2020; search in title)

4. "-BYT" "thông tin" (search date: March 5th 2020; search in title)

5. "-BYT" "thông minh" (search date: March 5th 2020; search in title)

6. "-BYT" "dữ liệu" (search date: March 5th 2020; search in title)

7. "-BYT" "trực tuyến" (search date: March 5th 2020; search in title)

8. "-BYT" "mạng" (search date: March 5th 2020; search in title)

9. "-BYT" "phần mềm" (search date: March 5th 2020; search in title)

10. "-BYT" "thiết bị" (search date: March 11th 2020; search in title)

*English explanation for search terms:

BYT: code for offical documents from the MoH (BYT is Vietnamese abbreviation of Ministry of Health); **điện tử** = electronic; **bệnh án** = health records; **công nghệ** = technology; **thông tin** = information; **thông minh** = smart; **dữ liệu** = data; **trực tuyến** = online; **mạng** = internet; **phần mềm** = software; **thiết bị** = devices.

Final search strategy for Pubmed search (performed on March 23rd 2020):

("Medical Informatics"[58] OR (health* AND informatic*) OR ((computerised OR digital OR electronic) AND health*) OR ehealth OR "e-health" OR (health* AND ("information system" OR information technolog* OR "information management system")) OR EHR OR EHRs OR EMR OR EMRs OR ((computerised OR electronic OR digital) AND (health record* OR patient record* OR medical record*))) AND (Vietnam OR "Viet Nam")

Web of Science search was conducted by a librarian at the Bodleian Health Care Libraries based on the final Pubmed search queries.

Multimedia Appendix 2

Circular 46/2018/TT-BYT on Regulations for electronic medical records

Content specifications of EMR

The content and classification specifications for records entered into EMR systems are based on the current regulations and MoH guidelines for medical records in Vietnam. Accordingly, EMRs are classified into inpatient record, outpatient record, and other specialized records. EMRs must be able to capture all the data that are normally collected in the MoH standard medical record templates. The relevant guidelines for EMR formats are shown below.

Decision 4069/2001/QD-BYT	Standard formats for medical records and documents.
Circular 50/2017/TT-BYT	Amendments for some regulations on healthcare cost payment.
Decision 4069/QD-BYT year 2010	Standard formats for traditional medicine records.
Decision 999/QD-BYT year 2011	Standard formats for abortion records.
Decision 3443/QD-BYT year 2011	Standard formats for ophthalmology records.
Decision 1456/QD-BYT year 2012	Standard formats for hand-foot-and-mouth disease records and some related regulations.

EMR creation and update

The maximum duration for updating information in an EMR system should be no longer than 12 hours from when the doctor gave their clinical decisions, or 24 hours if the clinical examination lasts longer than 12 hours or an IT issue occurs.

EMR storage and backup

The circular states that archiving paper medical records can be discontinued following EMR implementation if the following four criteria are met:

1. The EMR system has reached the “Advanced” level as defined in Circular 54/2017/TT-BYT (Assessment Criteria for Information Technology Implementation in Healthcare Facilities).
2. The system’s storage is capable of storing EMR data for the time span regulated in the healthcare law (10 years for outpatient/inpatient records; 15 years for accident records; and 20 years for mental disorders and mortality records).
3. Backup of EMR data is completed weekly and the backup data are saved at a data centre that satisfies the requirements from the Ministry of Information and Communication.
4. Organizations will transfer their EMR data to the new owner organization in case of mergers or dissolution.

Access right and secondary use

These regulations define the different groups that can request for access to information in EMRs, how the information can be accessed, and the restrictions for each group. These authorized groups were originally defined in the Healthcare Law, which regulates access to paper medical records. There are three groups with different levels of EMR access as follows:

- Staff of the healthcare facility, healthcare students, and researchers are able to view EMR data on-site or make an electronic copy of EMR contents for research or professional purposes.
- Representatives from the supervising agency of the healthcare facility, inspectors, procuracy, court officials, health inspectors, health insurance agencies, forensic agencies and lawyers can view EMR data on-site or make an electronic/printed copy of the EMR contents after being authorized by the head of the healthcare facility.
- The patient or their representatives can send a written request for their electronic or paper medical record summaries. EMR summaries must cover all the information listed in the standard medical record summary template (Appendix 4 of Circular 56/2017/TT-BYT on Detailed Guidance for Health-related Sections in Social Security Law and Hygiene and Labour Safety Law). Accordingly, information required in the medical record summary includes patient's name, demographics information, social insurance/health insurance ID, hospital admission/discharge information, and summaries of clinical progress, important test results, treatment, and discharge status.

All requests for access must be approved by the organization's authorities in charge. People who have authorization to access and use EMRs must ensure confidentiality and comply with the permitted rights.

Patient identification

Patient IDs follow the common national health ID system announced by the MoH.

Digital/electronic signatures

- Digital or electronic signatures are used to authenticate the health professional, the patient or their representatives who are responsible for the information on the EMR.
- When records are originally signed with electronic signatures, the healthcare facility must certify these signatures by an authorized digital signature.
- Organizations is required to develop and implement local policies on digital and electronic signatures in prior to their implementation.
- During the transition from paper-based medical records to EMRs, there are some paper documents that do not have a standard digital template and are unable to be signed with digital signatures. A workaround is allowed in which the document will be manually signed, scanned, and digitalized to be attached to the EMR profile, while the original document must be archived.

EMR functionalities

The EMR policy states that EMRs must comply with the following basic requirements:

- Use the MoH's terminology and service coding system and apply the health IT standards recommended in this circular.
- The EMR system must have user control capabilities including user authentication and authorization, an audit trail, and capabilities to protect confidentiality and security.
- The EMR system must be able export data to XML files customized to the following purposes:
    - Medical record summaries (as defined in Circular 56/2017/TT-BYT).
    - Primary health records (as defined in Decision 831/QD-BYT year 2017).

    - Sharing claim data to the health insurance portal (as defined in Decision 4210/QD-BYT year 2018).
- Data in EMRs can be displayed on computer screens or other devices
- Data in EMRs can be exported to the standard layouts of paper medical records and can be printed.
- Adhere to the functional modules defined in Circular 54/2017/TT-BYT (e.g., healthcare service management, administrative management, and medical record management).

Adherence to health IT standards

The circular recommends the EMR systems adopt specific health IT standards, including HL7 CDA, HL7 FHIR, DICOM version 2.0 or later, and relevant IT standards recommended by the Ministry of Health and the Ministry of Information and Communication.

Data security and confidentiality of EMRs

The circular requires that healthcare facilities have measures in place to safeguard the security and confidentiality of data stored in EMRs that suit the EMR system's specifications and organization's policies:

- The regulations on access rights and secondary use of EMR data described in this circular, and other relevant regulations from the MoH must be imposed in the facility.
- The facility must have solutions and protocols in place to protect cybersecurity during EMR implementation, including:
    - Access control solutions, including user authentication and authorization, and setting access time.
    - Preventing unauthorized access.
    - Restoring data lost in accidents.
    - Preventing, detecting and stopping malware.
- Data must be encrypted when being shared between facilities.
- Data must be encrypted based on relevant MoH guidelines.
- The EMR system must have audit trail functionality that can track data and time of activities such as viewing, submitting new data, editing, removing, and restoring data.
- Each facility must have their local policies and regulations on data security and patient privacy based on the current laws and guidelines.

Nomenclature system

The MoH's terminology and service coding system must be used to standardize data in the EMR system.

Criteria for LIS and PACS to discontinue physical film and paper-based archiving

The circular also sets out the criteria for LIS and PACS to discontinue keeping physical films and paper-based lab result registries. Accordingly, LIS and PACS implementation must reach Advanced level in Circular 54/2017/TT-BYT while their storage capacity can satisfy the archiving time similar to that of EMR systems.

Multimedia Appendix 3

Operational Requirements for Health IT Systems

Circular 53/2014/TT-BYT – Required conditions for provision of health IT activities

IT infrastructure

These criteria make sure healthcare facilities have appropriate infrastructure for efficient and secured service delivery. There are four domains covered: server and system software, network system, database and workstations. Specific requirements are presented in the table below.

Server and system software	Server infrastructure and associated equipment has sufficient performance, efficiency, data processing speed and data retrieval speed for the provision of online healthcare service
	The server has high availability and flexible backup mechanism for interrupted operation.
	The system software and operation system have legal license and transparent origin.
Network system	Network systems such as telecommunication network, Internet, wide area network, local area network and other connections are appropriately designed and implemented, with capable bandwidth. The telecommunication network must meet the requirements in Article 16 of Telecommunication Law.
	Network devices and network management software have legal license and transparent origin.
	Backup plan is pre-built to ensure network operation.
Database	Database for the online healthcare services must be stable and able to effectively process and store the data created during operation.
	Use a proprietary management system with a legal license and transparent origin, or open-source management system widely used nationally or worldwide.
Workstation	Decent number of workstations with appropriate configuration to operate the health IT services.

Information security

These are the requirements that help organizations to maintain system's security and protect their data from unauthorized access or accidents. Key solutions include cybersecurity policy, network security, application software security, data safety, and issue management protocols. Details for each solution are shown below.

Have SOPs for issue management	Have issue management SOPs that address personnel's responsibilities and detail solving steps including notifying the users and administrators. If the IT service is purchased externally, the provider must include these SOPs in the service package.
Issue check and management	Conduct frequent issue checks, update information about new issues and their management procedures
Cyberattack prevention	Employ techniques to timely detect and prevent cyber attacks
Proactive IT disaster prevention	Apply solutions to prevent IT risks and disasters to maximally limit cyber risks related to health IT activities

Human resource

Requirements for specialized IT staff in healthcare facilities are also addressed in this circular. These criteria examine professional competency, number of staff and routine trainings. In general, facilities at higher administrative levels need to meet higher standards.

Staff competency and quantity	Ensure there are enough IT specialized staff with competency.
Requirement for facilities at special level or level 1	These facilities must have IT department with at least 5 staff, of which at least 60% have Intermediate Degree or higher.
Requirement for facilities at level 2 or level 3	These facilities must at least have an IT group with at least 3 staff with Associate Degree or higher.
Training for health IT staff	Have training plans and organize trainings for health IT staff.
Hiring of external staff	Make sure the hired external staff have competency for the work. The contract must require adherence to healthcare law in protecting patients' privacy and confidentiality.

Requirements for specific heath IT services

In addition to the aforementioned IT conditions, health facilities need to follow particular standards and requirements during delivering their health IT services.

- Relevant procedures should be standardized so health IT applications can be implemented effectively.
- Health IT systems and applications should apply the following national and international standards:

HL7 standards (HL7 messaging 2.x, HL7 messaging 3.0, HL7 CDA)
DICOM (Digital Imaging and Communications in Medicine)
ISO/IEEE 11073 (a standard for medical device communication)
SDMX-HD (Indicators and metadata exchange standard)
IT standards recognized in Circular 22/2013/TT-BTTTT by the Ministry of Information and Communication

- Each healthcare organization must develop and enact their IT management and implementation policy.
- Secondary use of patient data must ensure patient's privacy and confidentiality regulated by healthcare law.
- Using digital signatures and digital identifications are allowed and regulated by Decree 26/2007/ND-CP, Decree 106/2011/ND-CP, Decree 170/2013/ND-CP, and Decree 106/2011/ND-CP.
- EMR creation, storage, and implementation must adhere to article 59 of healthcare law.
- When a healthcare facility employs an IT service from an external provider, the contracts and terms must address data ownership and relevant responsibilities when issues happen.

Multimedia Appendix 4

Circular 39/2017/TT-BTTTT - Technical Standards for IT Implementation in State Organizations

Type of standard	Standard code	Full name of standard	Application
Connection standards			
Hypertext transfer	HTTP v1.1	Hypertext Transfer Protocol version 1.1	Compulsory
	HTTP v2.0	Hypertext Transfer Protocol version 2.0	Recommended
File transfer	FTP	File Transfer Protocol	Compulsory application of one or both standards
	HTTP v1.1	Hypertext Transfer Protocol version 1.1	
	HTTP v2.0	Hypertext Transfer Protocol version 2.0	Recommended
	WebDAV	Web-based Distributed Authoring and Versioning	Recommended
Audio/ image streaming and transport	RTSP	Real-time Streaming Protocol	Recommended
	RTP	Real-time Transport Protocol	Recommended
	RTCP	Real-time Control Protocol	Recommended
Data access and sharing	OData v4	Open Data Protocol version 4.0	Recommended
Mail transfer	SMTP/ MIME	Simple Mail Transfer Protocol/Multipurpose Internet Mail Extensions	Compulsory
Provision of internet message access service	POP3	Post Office Protocol version 3	Compulsory application of both standards to the server
	IMAP 4rev1	Internet Message Access Protocol version 4 revision 1	
Directory access	LDAP v3	Lightweight Directory Access Protocol version 3	Compulsory
Domain name service	DNS	Domain Name System	Compulsory
Connection-oriented transport	TCP	Transmission Control Protocol	Compulsory
Connectionless transport	UDP	User Datagram Protocol	Compulsory
LAN/WAN internetwork	IPv4	Internet Protocol version 4	Compulsory
	IPv6	Internet Protocol version 6	Compulsory application of this standard to Internet-connected equipment
Wireless Local Area Network	IEEE 802.11g	Institute of Electrical and Electronics Engineers Standard (IEEE) 802.11g	Compulsory

	IEEE 802.11n	Institute of Electrical and Electronics Engineers Standard (IEEE) 802.11n	Recommended
Wireless Internet access	WAP v2.0	Wireless Application Protocol version 2.0	Compulsory
SOAP web service	SOAP v1.2	Simple Object Access Protocol version 1.2	Compulsory application of one, two or three standards
	WSDL V2.0	Web Services Description Language version 2.0	
	UDDI v3	Universal Description, Discovery and Integration version 3	
RESTful web service	RESTful web service	Representational state transfer	Recommended
Web services specifications	WS BPEL v2.0	Web Services Business Process Execution Language Version 2.0	Recommended
	WS-I Simple SOAP Binding Profile Version 1.0	Simple SOAP Binding Profile Version 1.0	Recommended
	WS- Federation v1.2	Web Services Federation Language Version 1.2	Recommended
	WS- Addressing v1.0	Web Services Addressing 1.0	Recommended
	WS-Coordination Version 1.2	Web Services Coordination Version 1.2	Recommended
	WS-Policy v1.2	Web Services Coordination Version 1.2	Recommended
	OASIS Web Services Business Activity Version 1.2	Web Services Business Activity Version 1.2	Recommended
	WS- Discovery Version 1.1	Web Services Dynamic Discovery Version 1.1	Recommended
	WS-MetadataExchange	Web Services Metadata Exchange	Recommended
Network time service	NTPv3	Network Time Protocol version 3	Compulsory application of one of the two standards
	NTPv4	Network Time Protocol version 4	
Data integration standards			
Extensible Markup Language	XML v1.0 (5th Edition)	Extensible Markup Language version 1.0 (5th Edition)	Compulsory application of one of the two standards
	XML v1.1 (2nd Edition)	Extensible Markup Language version 1.1	
Electronic Business Extensible Markup Language	ISO/TS 15000:2014	Electronic Business Extensible Markup Language (ebXML)	Compulsory
XML Schema	XML Schema V1.1	XML Schema version 1.1	Compulsory

Definition			
Data transformation	XSL	Extensible Stylesheet Language	Compulsory application of the latest version
Object modelling	UML v2.5	Unified Modelling Language version 2.5	Recommended
Resource description	RDF	Resource Description Framework	Recommended
	OWL	Web Ontology Language	Recommended
Character set demonstration	UTF-8	8-bit Universal Character Set (UES)/Unicode Transformation Format	Compulsory
Geographic information exchange format	GML v3.3	Geography Markup Language version 3.3	Compulsory
Geographic information access and update	WMS v1.3.0	OpenGIS Web Map Service version 1.3.0	Compulsory
	WFS v1.1.0	Web Feature Service version 1.1.0	Compulsory
XML metadata interchange specification	XMI v2.4.2	XML Metadata Interchange version 2.4.2	Recommended
Metadata registries (MDR)	ISO/IEC 11179:2015	Metadata registries - MDR	Recommended
Dublin Core metadata element set	ISO 15836- 1:2017	Dublin Core metadata element set	Recommended
JavaScript Object Notation (JSON) Data Interchange Format	JSON RFC 7159	JavaScript Object Notation	Recommended
Business Process Modelling Language	BPMN 2.0	Business Process Model and Notation version 2.0	Recommended
Information access standards			
Web content	HTML v4.01	Hypertext Markup Language version 4.01	Compulsory
	WCAG 2.0	W3C Web Content Accessibility Guidelines (WCAG) 2.0	Recommended
	HTML 5	Hypertext Markup Language version 5	Recommended
Extensible Web content	XHTML v1.1	Extensible Hypertext Markup Language version 1.1	Compulsory
User interface	CSS2	Cascading Style Sheets Language Level 2	Compulsory application of one of the three standards
	CSS3	Cascading Style Sheets Language Level 3	
	XSL	Extensible Stylesheet Language version	
Document	(.txt)	Plain Text (.txt) format: for unstructured documents	Compulsory

	(.rtf) v1.8, v1.9.1	Rich Text (.rtf) format v1.8, v1.9.1: for cross-platform document interchange	Compulsory
	(.docx)	Microsoft Word open document format (.docx)	Recommended
	(.pdf) v1.4, v1.5, v1.6, v1.7	Portable Document (.pdf) format v1.4, v1.5, v1.6, v1.7: for read-only documents	Compulsory application of one, two or three standards
	(.doc)	Microsoft Word document format (.doc)	
	(.odt) v1.2	Open Document Text (.odt) v1.2	
Spreadsheets	(.csv)	Comma separated Variable/Delimited (.csv) format: for exchanging data between different applications	Compulsory
	(.xlsx)	Microsoft Excel open XML spreadsheet file format (.docx)	Recommended
	(.xls)	Microsoft Excel spreadsheet file format (.xls)	Compulsory application of one or both standards
	(.ods) v1.2	Open Document Spreadsheets file format (.ods) v1.2	
Demonstration	(.htm)	Hypertext Document (.htm) format: for presentations exchanged through different types of browser	Compulsory
	(.pptx)	Microsoft Open PowerPoint file format (.pptx)	Recommended
	(.pdf)	Portable Document (.pdf) format: for read-only presentations	Compulsory application of one, two or three standards
	(.ppt)	Microsoft PowerPoint (.ppt) file format	
	(.odp) v1.2	Open Document Presentation (.odp) file format v1.2	
Graphic image	JPEG	Joint Photographic Expert Group (.jpg)	Compulsory application of one, two, three or four standards
	GIF v89a	Graphic Interchange (.gif) version 89a	
	TIFF	Tag Image File (.tif)	
	PNG	Portable Network Graphics (.png)	
Georeferenced image	GEO TIFF	Tagged Image File Format for GIS applications	Compulsory
Moving picture, audio	MPEG-1	Moving Picture Experts Group-1	Recommended
	MPEG-2	Moving Picture Experts Group-2	Recommended
	MPEG-4	Moving Picture Experts Group-4	Recommended
	MP3	MPEG-1 Audio Layer 3	Recommended
	AAC	Advanced Audio Coding	Recommended
Moving picture stream, audio stream	(.asf), (.wma), (.wmv)	Microsoft Windows Media Player formats (.asf), (.wma), (.wmv)	Recommended
	(.ra), (.rm), (.ram), (.rmm)	Real Audio/Real Video formats (.ra), (.rm), (.ram), (.rmm)	Recommended

	(.avi), (.mov), (.qt)	Apple Quicktime formats (.avi), (.mov), (.qt)	Recommended
Animation	GIF v89a	Graphic Interchange (.gif) version 89a	Recommended
	(.swf)	Macromedia Flash format (.swf)	Recommended
	(.swf)	Macromedia Shockwave format (.swf)	Recommended
	(.avi), (.qt), (.mov)	Apple Quicktime formats (.avi),(.qt), (.mov)	Recommended
Mobile content	WML v2.0	Wireless Markup Language version 2.0	Compulsory
Character set and encoding	ASCII	American Standard Code for Information Interchange	Compulsory
Vietnamese character set	TCVN 6909:2001	TCVN 6909:2001 "Information Technology - 16-bit Coded Vietnamese Character Set"	Compulsory
Data compression	Zip	Zip (.zip)	Compulsory application of one or both standards
	.gz v4.3	GNU Zip (.gz) version 4.3	
Client-side scripting language	ECMA 262	ECMAScript version 6 (6th Edition)	Compulsory
Web content sharing	RSS v1.0	RDF Site Summary version 1.0	Compulsory application of one of the two standards
	RSS v2.0	Really Simple Syndication version 2.0	
	ATOM v1.0	ATOM version 1.0	Recommended
Network Protocol Standard	JSR 168	Java Specification Requests 168 (Portlet Specification)	Compulsory
	JSR286	Java Specification Requests 286 (Portlet Specification)	Recommended
	WSRP v1.0	Web Services for Remote Portlets version 1.0	Compulsory
	WSRP v2.0	Web Services for Remote Portlets version 2.0	Recommended
Information security standards			
Email security	S/MIME v3.2	Secure Multi-purpose Internet Mail Extensions version 3.2	Compulsory
	OpenPGP	OpenPGP	Recommended
Transport layer security	SSH v2.0	Secure Shell version 2.0	Compulsory
	TLS v1.2	Transport Layer Security version 1.2	Compulsory
File transfer security	HTTPS	Hypertext Transfer Protocol Secure	Compulsory
	FTPS	File Transfer Protocol Secure	Recommended
	SFTP	SSH File Transfer Protocol	Recommended
Mail transfer security	SMTPS	Simple Mail Transfer Protocol Secure	Compulsory
Message access service security	POP3S	Post Office Protocol version 3 Secure	Compulsory application of one or both standards
	IMAPS	Internet Message Access Protocol Secure	

DNS security	DNSSEC	Domain Name System Security Extenssions	Recommended
Network layer security	IPsec - IP ESP	Internet Protocol security với IP ESP	Compulsory
Wireless network security	WPA2	Wi-fi Protected Access 2	Compulsory
Encryption algorithm	TCVN 7816:2007	Cryptographic technique - Cryptographic algorithms - Data Encryption Algorithm AES	Recommended
	3DES	Triple Data Encryption Standard	Recommended
	PKCS #1 V2.2	RSA Cryptography Standard - version 2.2	Recommended, using RSAES-OAEP scheme for encryption
	ECC	Elliptic Curve Cryptography	Recommended
Digital signature algorithm	PKCS #1 V2.2	RSA Cryptography Standard - version 2.2	Compulsory, using RSASSA-PSS scheme for signature
	ECDSA	Elliptic Curve Digital Signature Algorithm	Recommended
Digital signature hash algorithm	SHA-2	Secure Hash Algorithms-2	Recommended
Key transport algorithm	RSA-KEM	Rivest-Shamir-Adleman - KEM (Key Encapsulation Mechanism) Key Transport Algorithm	Compulsory
	ECDHE	Elliptic Curve Diffie Hellman Ephemeral	Recommended
User authentication solution	SAML v2.0	Security Assertion Markup Language version 2.0	Recommended
XML message exchange security	XML Encryption Syntax and Processing	XML Encryption Syntax and Processing	Compulsory
	XML Signature Syntax and Processing	XML Signature Syntax and Processing	Compulsory
XML public key management	XKMS v2.0	XML Key Management Specification version 2.0	Recommended
Personal information security protocol	P3P v1.1	Platform for Privacy Preferences Project version 1.1	Recommended
Public key infrastructure			
Cryptographic message syntax for signing and encrypting	PKCS#7 v1.5 (RFC 2315)	Cryptographic message syntax for file-based signing and encrypting version 1.5	Recommended

Cryptographic token information syntax	PKCS#15 v1.1	Cryptographic token information syntax version 1.1	
Private-key information syntax	PKCS#8 V1.2 (RFC 5958)	Private-Key Information Syntax Standard version 1.2	
Cryptographic token interface	PKCS#11 v2.20	Cryptographic token interface standard version 2.20	
Personal information exchange syntax	PKCS#12 v1.1	Personal Information Exchange Syntax version 1.1	
Certificate revocation list format	RFC 5280	Certificate Revocation List Profile	
Digital certificate format	RFC 5280	Public Key Infrastructure Certificate	
Certification request syntax	PKCS#10 v1.7 (RFC 2986)	Certification Request Syntax Specification version 1.7	
On-line Certificate status protocol	RFC 6960	On-line Certificate status protocol	
Time stamping protocol	RFC 3161	Time stamping protocol	
Time stamping services	ISO/EEC 18014-1:2008	Information technology Security techniques - Time stamping services	
	ISO/EEC 18014-2:2009	Part 1: Framework	
	ISO/EEC 18014-3:2009	Part 2: Mechanisms producing independent tokens	
	ISO/EEC 18014-4:2015	Part 3: Mechanisms producing linked tokens	
		Part 4: Traceability of time sources	
Web services security	WS-Security v1.1.1	Web Services Security: SOAP Message Security Version 1.1.1	Recommended
Incident object description exchange format	RFC 7970	The Incident Object Description Exchange Format version 2 (IODEF)	Recommended

Multimedia Appendix 5

Circular 48/2017/TT-BYT - Regulations on data exchange in management and reimbursement of health insurance claims

There are two portals that organizations need to submit their data to, the health insurance portal from the VSS and the health data portal from the MoH. While the VSS's portal serves the health insurance claiming and investigation purposes, data submitted to the MoH's portal will be stored in the national database for research and policy making purposes. A healthcare facility will submit the same dataset to both portals following a set of guidance for data sharing and formatting.

Requirements for data format

The circular categorizes data used for health insurance claiming purpose into two groups, input and output data. Input data is the data collected and stored in the IT system while output data is compiled and exported from the IT system for sharing purposes. Specifications for these data are as followed.

Data coding and nomenclatures	The terminology and service coding systems of the MoH is required for both input and output data
Data format	Data must be formatted in XML format and uses the UTF-8 encoding system
	One XML file can contain one or multiple health packages containing the information of one visit bout, even when the client has multiple insurance IDs

Data sharing protocols

The data can be shared to the portals through one of these protocols

- Protocol 1: via the web service.
- Protocol 2: data sync from the software installed on a workstation computer.
- Protocol 3: direct data import.
- Protocol 4: using the File Transfer Protocol (FTP).

In either of these protocols, healthcare facilities must ensure data integrity is maintained.

Instant data sharing and feedbacks

Healthcare facilities submit the data to the health insurance portal "instantly" after the end of a doctor's visit, an outpatient treatment period, or an inpatient discharge. These data do not need to go through electronic verification.

Once received the data, the health insurance portal must reply the sender with a confirmation of receiving data along with other information that is:

- Information about the current status of the health insurance account.
- The patient's medical history for the last six months, including:
  - Time points of the last visits.
  - The primary diagnoses and comorbidities classified with ICD-10 or the traditional

medicine classification system.

Reviewed data sharing and feedbacks

A secondary data submission is carried out within seven working days after ending a healthcare activity or a treatment period. The data that have been transferred instantly will be reviewed, corrected for any discrepancies, and electronically verified. Data will then be submitted to two portals, the health insurance portal and the health data portal. If the data is generated in the end of a month/quarter/year, data submission should be conducted by the fifth day of the next month.

The portals must notify the sender if data is successfully received. The health insurance agency will assess the claim information and reply to the healthcare facilities in details about the investigation results within seven working days since data arrival. Conflicts and queries related to the claim, if any, must be clearly indicated in the relevant data field of the XML file to facilitate effective communication and timely justification between the healthcare facilities and the health insurance agency.

Delay in data sharing

Delay in data sharing is acceptable in some particular cases. Following is the policies related to sharing delay.

Cases when delay in data sharing is acceptable	When there are unpredictable technical errors of the information systems that obscure data transfer and feedbacks
	When there is an interruption of power supply or Internet connection, or incapable IT infrastructure that impedes data transfer and feedback
Notification of the interruption	Healthcare facilities must notify the portal admins about the issues, and vice versa via phones, emails or faxes
	Data transfer and feedback must be continued after the issues have been resolved
Provide alternative communication solutions when IT infrastructure is incapable	The healthcare facility and the VSS must agree on the alternative solutions for data transfer and address them clearly in contract then inform the supervising organization about the substitution

Data security and data management

The data exchange activities and the data itself are protected and managed in compliance with relevant cybersecurity regulations and healthcare regulations.

Organizations engaging in the data exchange activities must be proactive and coordinative, within permitted responsibilities, to ensure safety, security, accuracy, and integrity of the data.

Multimedia Appendix 6

Decision 4159/QD-BYT year 2014 - Guidance on ensuring security of electronic health data in health organizations

Organization's information security policies

Organizations must have local information security policies for their IT system. The policies need to be approved by the organization's management board and compliant with relevant regulations of the government. The policy's structure should at least cover the following sections:

- Background (scope, applicable groups, concepts, aims).
- Detail policies (key points, standards, specific requirements).
- Roles and responsibilities of relevant stakeholders.
- Policy implementation.

The policies should be annually reviewed and modified if needed to ensure its appropriateness, adequacy and efficiency.

Intranet and Internet security

Organizations must apply measures to safeguard their intranet and Internet security, including:

- Solutions to detect and prevent intrusion and malwares.
- DoS prevention.
- Verification solutions for wireless network.
- Network partition measures to manage access and provide quick and efficient access.
- Backup for important components in the system.
- Demand evaluation to ensure suitable bandwidth for the local and Internet network.
- Regular updates for the network and security devices.
- Ensuring an adequate supply of trustworthy security and network infrastructure for the organization.

Server and system software

The requirements for server and system software are shown in the table below:

Server infrastructure	The server and related equipment have good performance and efficiency for implementing IT activities.
Requirements for server	High availability and flexible back-up mechanism.
	Having separate and suitable server room, strict protection and supervising of server room, suitable physical environment e.g. power supply and room temperature for uninterrupted operation, fire safety solutions, and assigning clear roles and authorization for personnel having access to the room.
Remote access, security, and working environment for the server	On-site or remote access to the server must be protected with passwords or other means of control; server should be assigned to a separate and proper network partition.
Intrusion and virus prevention	Have protocols to detect and prevent intrusion, malwares and virus for server

Requirements for system software	Operating system installed on server have legal license or are popular open-source OS such as UNIX, LINUX
	System software have technical documents
	Regular checks and updates of software patches

Workstation computers

Workstation computers in the organizations need to be protected from a variety of threats including unauthorized access, viruses, confidentiality breach, and system errors:

- All workstation computers including desktops and laptops must be password-protected and/or secured with other measures.
- Databases or folders keeping important health information must be password-protected.
- Organizations must have protocols to detect and prevent workstation computers from intrusion, malwares, and viruses.
- Organizations must implement measures to protect data in workstation computers if they are connected to the Internet.
- Workstation computers at places where communication with customers happens need to be prevented from unwanted view of confidential information. Screen lock when not in use setting should be turned on.
- Frequent updates of error fixes and upgrades of the operating system.

Application software

Application software security set out responsibilities of the software provider, the supporting agency and the host organization in preserving cybersecurity related to application software.

Security requirements from the manufacturer/provider	Security requirements from the manufacturer/provider need to be clearly addressed in the technical document. These requirements should be thoroughly adhered to during using the software
Risk assessment and prevention	Risks should be assessed, and risk prevention measures should be continuously conducted before the software is installed until it is in use
Security testing and assessment	Before accepting the software, relevant security testing and assessment must be conducted and documented. The test should take place in a separate environment not affecting the organization's ongoing activities
Account management	Strict management of all the user accounts
Version management and updates	Evaluate efficiency, effect and risks when integrating with other software and when upgrading to a new version
	New versions should experience security testing and earn authority's approval before being implemented
	Have recovery protocols if issues occur during software upgrade
	The upgrade procedure should be documented with relevant purpose, version, time, person conducting the upgrade
	Software versions should be well managed and stored in a secured place
	The upgrade procedure should be supported with detailed guiding documents

Management of source program	Assign personnel to manage the software's source program
	Access to source program must be authorized and documented
	Source program must be kept safely at two separate locations
	The software provider and the supporting agency must provide assurance of malware-free to the customer
Software's license	Encourage using licensed software, strictly avoid using illegal license. The organization or individuals must be responsible for consequences of using illegal software

Email

Email security requires good email management from the organization as well as safe ownership and practice from the users:

- Using public email addresses to exchange work emails should be avoided while work email addresses should not be used for personal purposes.
- Regular password changing and using non-editable document formats should be encouraged.
- Emails should only be used by the authorized users. Passwords for emails should have good complexity and can only be changed by the authorized person.
- Staff's emails should be locked or deleted when they stop working for the organization.
- The organization must have measures to ensure the email system's security and readiness, as well as junk mail prevention measures in the intranet and Internet environment.

Database

- Organizations must only use database management systems (DBMS) with legal license and origin, or popular open-source DBMS such as MySQL, PostgreSQL, MongoDB.
- DBMS need to satisfy the following criteria: stability, being able to store and handle the organization's data, having measures to protect and authorize access to the DBMS resources.
- Regular updates to fix DBMS's errors.
- Organizations should develop backup and recovery protocols for the database.
- Organizations should implement user authorization with clear assignment of roles for individuals accessing the database. Logging activities on the database is encouraged but this should not interfere with data processing speed.
- Protocols to prevent intrusion and unauthorized access should be implemented.

Backup and recovery

The data backup and recovery requirements below cover workstation computers, servers, and software in an organization.

Data on workstation computers	Important data should be backed up right after modification to allow for recovery if needed
	Important data needs to be backed up before changes or upgrades in the operating system are made
	Backup devices and recovery procedures should be checked regularly to ensure preparedness
	Backup data must be stored in a safe location separated from the original data and unauthorized personnel. Important data should ultimately be stored separated from the host facility's geographic location

Data on servers	Backup data and data recovery documents must be stored in a location separated from the installation location to avoid severe issues. Backup frequency, especially of important data, must be planned to meet the organization’s recovery needs
	Backup devices should be regularly checked to ensure preparedness
	Backup data should be stored at a location that has physical protection and meets the same protection standards applied for the main storage place
	Storing duration for backups of important information and requirements for permanent backups should be identified
	Database recovery should be regularly checked to ensure efficiency and capability to recover data within the time limit
Software	Original version of the software must be safely archived for quick reinstallation if needed

Data transmission

Health facilities should use identification and encryption methods to protect their data during data exchange. Specific requirements in using these methods are as followed:

- Identification methods allowed by Vietnam law and the MoH should be used in data transmission.
- Using the encryption methods suitable for the security configuration and processing capacity of the information system.
- The encryption keys must be created, modified, distributed, and stored in a safe manner.
- Ensuring the encrypted data can be decrypted when needed.
- Developing policies about withdrawing, eliminating, and restoring encryption keys.

User accounts

This section addresses the good practice in user account management that organizations need to follow. Details are presented below.

Access authorization policies	Developing organization’s policies for authorizing access right to network, server, application software, and database, covering from access requesting to access removal
Privileged account management	Developing policies in controlling and supervising access to the privileged accounts
Security requirements in user account management	Each user is only allowed to own one account
	Defining access groups and authorizing accounts based on these groups
	Passwords should be changed routinely (at least once a month)
	The organization should have definitions for a strong password such as the limited character counts, required availability of both upper case and lower cases, or both alphabet and numeric characters
	A temporary password is created for a new account and will be changed after first sign-in. If possible, no reuse of old passwords should be applied
	Have procedures to instantly remove accounts and halt authorizations of staff who change their role or stop working

	for the organization

Remote access

To secure remote access activities, organizations must make sure the following instructions are followed:

- User authentication and recognition capability of the security system:
  - Able to proactively identify the user and their access right.
  - At minimum, passwords are used to recognize users.
  - Important resources can be protected by more advanced solutions such as smart cards, token keys, and biometrics.
- Protection for data during transmission: encryption methods can be used to protect important data during transmission.
- Protection for resources in the network: implementing protocols for controlling resources subject to remote access.

Elimination of storage devices

Storage devices intended to be eliminated need to be handled appropriately to avoid unwanted data security issues.

- Before being eliminated, devices storing important health data need to be checked and ensured that any data or licensed software is removed or re-formatted.
- Storage devices no longer able to work but used to stored important data must be physically destroyed before elimination.

Ensuring continuity of the IT system

Uninterrupted operation of the IT system is an important objective that healthcare facilities need to achieve. Below is the recommended measures to maintain IT systems' continuity:

- Developing and conducting plans to maintain continuity of the information system.
- Building data backup and recovery protocols.
- Ensuring data can be access quickly and uninterruptedly.
- Having network backup solutions.
- Having protocols to ensure continuity of the server system. Using technologies to support server availability is encouraged.

Issue management

Potential cybersecurity issues need to be prevented through appropriate protocols and techniques. Recommended measures are shown below:

- Organizations should develop internal IT issue management procedures. The procedures should be reviewed and updated with new issues and solutions at least every six months.
- Employing techniques to detect and solve DoS attacks such as firewall devices, intrusion detection and prevention devices, attack alert devices, and packet filter.
- Organizations should request the service providers to provide issue management protocols for their services.

Human resource to ensure cybersecurity

To ensure the system's cybersecurity, organizations is required to the following guidance:

- The organization should assign at least one person with appropriate competency to the cybersecurity position.
- System administration, application software development and maintenance, and system operation tasks should be assigned to specific teams and individuals. No individuals should have the full authorization for the system except ones approved by the organization's head. Relevant responsibilities of these groups and individuals should be clearly addressed in the organization's policy.
- Publicizing policies for strict management of system access on privileged accounts.
- Organizing trainings and continuous education for cybersecurity staff.
- Recruited staff should be educated about organization's information safety policies.

Supervising external IT providers and supporters

Technicians from service providers or technical services who temporarily work with the organization's IT system need to be supervised to avoid unwanted cybersecurity issues. In particular, organizations should develop and implement thorough protocols for supervising external technicians working with the IT system. These protocols must be approved by the organization's authority.

Multimedia Appendix 7

Laboratory Information Systems
Decision 3725/QD-BYT year 2017 - Guideline for implementation of laboratory information system (LIS) in healthcare facilities

Information for new lab test orders

LIS should be able to record the information of new orders by three groups, patient's personal information, sample information, and order information. Data items for each group are presented below.

Patient's personal information	Patient ID, administrative information such as name, age, sex, address, department, room, bed, and health insurance ID
Sample information	Sample code, type of sample, sample source, department ordering test, and sample receiving time
Test order information	Service code, service name, department and staff odering test; test order by individual test or by package of tests; capability to add test order to an existing sample

Data entry methods

Data can be transferred to LIS via multiple methods such as:

- Laboratory staff manually enter data to LIS using the data entry interface.
- Information is sent from HIS to LIS.
- Information is read to LIS by scanning the codes (barcode, QR code, other types of code) printed on test referral sheets.

Sample labelling

LIS can create unique sample IDs that are used throughout the lab procedures. Information on sample labels must have sample IDs at minimum.

Order status and history viewing

LIS enables users to search, view, and print information related to a patient or a sample. The viewed information can be sample's ID, patient name, sample status, time points, and test history.

LIS can support testing management tasks via managing lists of orders and related details and results. Progress of an order can be shown on LIS e.g. sample taken/not taken, sample processed/not processed, results available/not available.

Queue management

LIS has capability to manage queues of sample collection and queues of pending results. These queues can be displayed on screens to inform patients.

Result management

LIS's result management capability involves the essential functions such as updating, archiving, displaying, summarizing, and printing test results as well as the extra functions e.g. alert, writing notes, and locking records. Details of these functions are as followed.

Test and administrative information update, display, and archive functionality	Patient's administrative and clinical information
	Information about the department and the doctor making the order
	Information of the result: test name; date and time of ordering, collecting, and analyzing sample; test results by various indicators; time of announcing result; data entry person and other relevant information
	LIS can integrate with lab machines to directly receive the results
Alert for abnormal results	
Allowing users to write notes for the results (if necessary)	
Locking results after completion	
Managing and summarizing test results based on the standard templates in Decision 4069/2001/QD-BYT	Templates in Decision 4069/2001/QD-BYT: - Lab test voucher (27/BV-01) - Hematology result sheet (28/BV-01) - Template 29/BV-01 - Template 30/BV-01 - Template 31/BV-01 - Template 32/BV-01 - Template 33/BV-01 - Template 34/BV-01 - Template 35/BV-01 - Template 36/BV-01 - Template 37/BV-01
Printing test results and export result documents to popular file formats	Some popular digital formats such as .doc, .docx, .xls, .xlsx, .pdf

Inventory management

This functionality helps staff to manage and monitor the laboratory chemicals and supplies, including:

- Inventory management: reporting the inventory, generating warehouse input/output certifications, and inventory viewing.
- Keeping track of the supplies and chemicals taken from the inventory to run a test.
- Setting consumable standards of supplies and chemicals for each type of test.
- Summarizing the amount of supplies and chemicals used for each type of test.
- Summarizing the amount of supplies and chemicals used in each test device.
- Managing, summarizing and printing inventory reports.
- Summarizing supplies and chemicals by patch number, expiring dates. LIS can prioritize supplies and chemicals with the nearest expire dates.

Interoperability between LIS and lab machines

- LIS is able to integrate with lab machines to send orders and receive results automatically. In particular, there are two types of communication between LIS and lab machines:
    - In one-way communication, LIS can automatically receive results from the lab machines.

  - In two-way communication, LIS is not only able to automatically receive results from the lab machines, but also able to send orders and commands to the lab machines.
- LIS can define and automate the mapping process for lab test names for standardization.
- LIS can receive and read graphs from some lab machines such as hematology analyzers, electrophoresis equipment, and electrocardiograph machines, or can generate graphs from the data extracted from these machines.
- LIS can connect with the devices that can connect with popular lab machines in Vietnam via the standards such as RS232, RJ11, RJ45, USB.
- LIS can receive and store the images captured from the devices connected to it.

Interoperability between LIS and HIS

- LIS receives patient information and order information from HIS instead of manually entering these information to LIS; results are transferred from LIS to HIS
- LIS supports sending test orders to lab machines having two-way connection.
- LIS can exchange data with other laboratories and with other information systems to support management and administration purposes.

Reporting and statistics

Organizations can use LIS to produce summaries and reports to their managers, the MoH, or the VSS. Following is all types of reports that LIS should be able to generate.

Generating summaries from specific lists	List of received samples
	List of samples to collect
	List of pending tests
	Summaries for all tests
	Chemical quotas
Managing and summarizing data, and exporting reports based on the registry formats in Decision 4069/2001/QD-BYT	The relevant registry formats in Decision 4069/2001/QD-BYT are: • The lab registry (14/BV-01) • The peripheral blood cell test registry (15/BV-01) • The microbiology registry (18/BV-01) • The test result announcement form (20/BV-01)
Generating summaries and reports customized to the host organization and the VSS' requests	

Implementing terminology and service coding systems

The MoH requires healthcare facilities to widely adopt the MoH terminology and service coding system when implementing LIS to create consistency between technical procedures, administration, and reporting. Below is the terminology and service coding lists that LIS are required to adopt:

- Lists of lab test indicators, lists of lab test indicator groups; lists of lab service categories; lists of lab techniques, lists of technical service categories; lists of lab sample categories; lists of lab instrument; lists of medical supplies, lists of blood and blood products (published by the MoH).
- Lists of hematology-transfusion, biochemistry and microbiology techniques in Circular 43/2013/TT-BYT and its amendments in Circular 21/2017/TT-BYT.

- LOINC standard (recommended in Decision 2035/QD-BYT year 2013).

In addition, LIS can be updated with extra terminology systems or newer versions of the installed systems.

System administration

The system administration functionality group includes the following functions:

User administration	LIS user administration
	Authorization
	Audit management
Setting alert threshold for abnormal results	
LIS configuration	Setting configurations for database connection
	Setting configurations for work modes
	Backup
	Audit mode
	Log in, log out and other related functions

IT infrastructure and human resource to operate LIS

<u>IT infrastructure</u>

IT infrastructure conditions to implement LIS need to satisfy the relevant criteria addressed in Article 3 of Circular 53/2014/TT-BYT (Required conditions for provision of health IT activities). In particular, capacity of the network, server, and auxiliary devices to operate LIS must be ensured as followed:

- The local network is properly designed and implemented, with proper bandwidth.
- The server system should have appropriate performance ability and efficiency to operate the lab system. High availability and flexible backup mechanism is also required for the server system to operate LIS without interruptions.
- Auxiliary devices such as workstation computers and printers should be installed with suitable quantity and capability to cooperate with LIS and other systems.

<u>Cybersecurity</u>

Measures to protect cybersecurity during implementing LIS should seek to meet the requirements in Article 4 of Circular 53/2014/TT-BYT (Required conditions for provision of health IT activities) and Decision 4159/QD-BYT year 2014 (Guidance on ensuring security of electronic health data in health organizations).

<u>IT workforce</u>

At minimum, a facility must have one IT specialized staff with an associate degree or higher to administer and operate LIS.

Multimedia Appendix 8

Hospital Management System
Decision 5573/QD-BYT year 2006 – Requirements and functional modules for hospital management software

<u>General requirements</u>

HMS development and implementation must be based on the current laws and regulations of the government and the MoH. The specific requirements are given as followed:

- HMS development and implementation must comply with the current laws from the government.
- The HMS must satisfy all the hospital administrative tasks regulated by the MoH.
- Data and template configuration used in HMS must ensure appropriateness with data structure of the reporting and medical record system of the MoH.
- The MoH Terminology and Service Coding System should be universally used in the facility to maintain consistency throughout healthcare delivery, reporting and payment. Double data entry should be avoided.
- Following MoH requirements and standards in reporting and managing medical records.

<u>Technical requirements</u>

- The HMS can communicate with the Medisoft 2003 software, or can export data and generate reports that satisfy Medisoft 2003's report standards.
- The HMS can communicate with the Health Insurance software, or can export data and generate reports requested by the Vietnam Social Security.
- The facility must have solutions to maintain security of HMS including ensuring data safety and information security, having user authorization and authentication, and capability to strictly control user activities and prevent unauthorized access. The security system must have at least 3 layers: system, database, and user.
- Using UTF-8 for character encoding.
- Requirements for operation system, database and software development language:
    - The development of HMS that can run on free operation systems and database management systems are encouraged.
    - HMS has data backup and recovery functionality.
    - Having solutions for the database management issue.
    - Legal license of the development language can be proved.
    - Maintaining objectivity and honesty between database and reports.
- The software's design is openness-oriented and easy for troubleshooting, maintenance and upgrading.
- Advanced technologies and techniques are encouraged in developing and implementing HMS.
- Some nomenclature systems and health IT standards recommended for HMS are listed below:

Nomenclature systems	
System	**Author organization**
The List of Administrative Units of Vietnam	Vietnam Government

The Hospital IDs List	MoH
The First-line Healthcare Facility IDs for Health Insurance Users	VSS
The Anatomical Therapeutic Chemical Classification System (ATC)	WHO
WHONET	WHO
ICD-10	WHO
Health IT standards	
System	**Author organization**
HL7 messaging*	HL7
Digital Imaging and Communications in Medicine (DICOM)	
Picture Archiving and Communications Systems (PACS)	

*If HL7 standards are currently not supported in the HMS, the provider must commit on providing technical documents or supports to help the hospital's system connect with other information systems of the hospital, or the Department of Health and the MoH systems.

HMS functional modules

To adequately address the administrative tasks in a hospital, an HMS need to be incorporated with a wide range of functionalities. This guideline presents eight modules that an HMS should have, of which the constituent functionalities are explained as followed.

Outpatient management module

The purpose of this module is to manage all administrative and health service data of the patients. These data can be accessed in the whole hospital, and is used at the next patient's visits. Key functionalities and instructions are provided below:

Registration management	Generate and distribute unique patient IDs which will be used in the next time the patient visits
Manage the data recorded by the MoH standard medical record template	Demographics data: fullname, date of birth/age, 4 levels of address: village/house number - commune/ward/street - district - province/city
	Patient group data: cost-free, pay, health insurance, poor group, children under 6 years old, and other groups
	Health insurance related data (regulated by the VSS): health insurance ID, first-line healthcare facility, valid date, beneficiary code, first health insurance provider, reasons of care seeking
	Referral place info: ID, facility's name etc.
Room management	Diagnosis management by ICD-10 (4 characters): medical history, previous diagnosis, primary diagnosis, other diagnosis

	Doctor visit management: date & time of examination, doctor's name, data entry staff's name
	Order management (lab tests, medical imaging, treatment)
	Prescription management, including archiving and printing
	Manage doctor's decisions: outpatient treatment, admission, clinic referrals
	Print the MoH examination sheet for patients admitted to hospitals
Outpatient management	Outpatient medical record management
	Outpatient order and service management
	Statistics for outpatient duration
Manage patients monitored at the outpatient ward	Medical professional management
	Service management
Outpatient test management	
Pharmacy management	

Inpatient management module

This module manages data of the patients admitted to the inpatient departments. Data collected includes administrative data, diagnoses, room management, and surgery management. Details of each functionality are as followed.

Patient information management	All administrative information in the admission sheet and the medical record template from the MoH.
Disease information management	Adopt the ICD-10 with 4-digits coding.
	Diagnosis management: previous diagnosis, outpatient diagnosis, primary diagnosis, accompanied diagnosis, medical history, main ward admission diagnosis, main ward referral diagnosis, main diagnosis at discharge, diagnosis at death, and diagnosis at autopsy.
Ward and room management	Bed management: number of beds, bed types, fees, tracking occupation of each bed type.
	Ward discharge and referral to another ward.
	Hospital discharge and referral to another hospital.

Surgery management	Organize surgery schedule: patient name, surgery time, main surgeon, main anaesthetist, and other surgery attendants; surgery types and cost.
	Manage information recorded in the surgery info sheet and surgery outcomes.
Reporting and statistics	Summarize and export treatment reports by the 11 hospital reporting templates.
	Generate reports under requests from the VSS and other agencies (if necessary).
	Generate reports under requests from the host hospital.

Medical test management module

It is noted that this module can be implemented by phases that reflect the hospital's resource and information infrastructure. Priority should be given to order management (for summary and cost calculation), followed by result management (for clinical practice and electronic health records) and communicating with lab machines. This is considered a complex module that requires integration with other modules such as outpatient management, inpatient management, pharmacy and inventory management, and financial management. More guidance is presented below.

Use the terminology and coding system from the MoH to manage medical tests	
Outpatient test management	Manage test orders from the outpatient ward, including patient ID, patient name, name of clinic making order, name of test, order time, name of doctor making order
Inpatient test management	Manage orders by patient
	Manage results by patient
	Complications occurred during a procedure
	Manage and exchange data collected from the test machines (if available), such as images, videos, audio
Result management at the lab test and medical imaging departments	Manage and store all the results available at the department, together with the patient information
	Manage administrative and clinical information
	Manage information related to the ward and personnel making orders
	Result information: name of test, ordering time, sampling time, sample processing time, name of the person conducting the test, test result, result announcement time, name of data entry staff etc.
	Interoperability with test machines for direct result receipt should be aimed at.

Test cost management	
Supply and chemicals management	
Reporting and data export	Reports for testing activities can be generated under hospital's regulated templates.
	Can generate reports requested by the hospital

Pharmacy management module

The pharmacy management module aims to provide effective and safe management of the drug inventory in the hospital. Following is the key functionalities that this module entails.

Drug and supply management	Have a standard terminology system for drugs and supplies applicable for the whole hospital.
	Manage drug expired dates, utilizing a tool to monitor the expired dates and alert to drug expiration.
	Able to withdraw drugs under the requests from the Pharmacy Administration Agency.
	Have a pharmacopoeia to guide drug prescribing.
Inventory management	Develop catalogs to manage drug inventory import and export. Build a platform to create, edit, and remove the catalogs.
	Have a data entry UI that can track inventory input by sources such as from providers, returned from the wards, produced internally, and record related information required by the MoH.
	Have a data entry UI that can track types of inventory output such as return to providers and small-amount export, and record related information required by the MoH.
	Have a data entry functionality to add products used for other output purposes such as disease prevention, scrapping, and liquidation. For each of these cases, design a suitable function to verify output, including generating output certificates and output vouchers.
Drug delivery management	Have a platform to manage drug delivery based on prescriptions, in which patients with health insurance and outpatient patients can be distinguished. In some cases, copies of outpatient prescriptions can be created.
	The module can anticipate the needed drugs based on electronic data of medical records. Can differentiate between reserved by treatment demands and drugs reserved by default.

	Have a functionality to return drugs and make related verification processes when the patient dies, changes drugs or withdraws from treatment.

The guideline also set specific requirements for the module as well as the inventory management practice:

- The import and export process must adhere to the first in first out method, and be based on drug expiration dates as well as other regulations on keeping and delivering drugs.
- The module is able to manage inventory input-output and drug delivery by financial sources such as public funding, health insurance funding, and charity funding.
- The module has capability to generate quick and accurate summaries and reports of import, export and balance of inventory.
- The module has search functionality based on various criteria.
- The printable templates and registries in the module must adhere to the relevant pharmacy regulations.
- The module is able to generate reports of hospital pharmacy activities under the formats required by the MoH, and other reports requested from the Department of Health and the hospital.

Hospital fee and health insurance management module

This module is considered a crucial component of a HMS. It communicates with other HMS's modules, and is installed at the finance department and payment points in the hospital. Hospitals employing HMS should not use other software to manage hospital fees and health insurance to avoid double data entry and waste of resources. Detailed guidelines for the module's functionalities are provided below.

Adopt a unique service terminology system in management	Use the terminology and service coding system from the MoH.
	Manage prices of treatment services based on current regulations from the MoH and the VSS.
Publicize costs for patients	Able to calculate hospital costs for any patient, at any point of time, under any payment protocol.
Manage costs and payments by specific groups	Direct payment group: self-pay patients, poor patients, autonomic finance management.
	Indirect payment group: health insurance-covered groups (whole-cover and partial-cover), under-6 children.
	Free group.
	Other groups.
Outpatient cost management	Examination fee, medical test fee, surgery fee, treatment fee.
Inpatient cost management	Management of advance fee (for direct payment group).
	Management of treatment costs, including medication, blood and fluid transfusion; bed fee; surgery; medical tests.
	Publicize everyday costs: able to calculate patients' treatment

	costs anytime.
	Able to print the standardized and customized invoices.
	In health insurance reimbursement management, the module is able to distinguish between costs covered and not covered by health insurance.
Hospital cost management for patients with health insurance	Able to distinguish between costs covered and not covered by health insurance.
	The software can generate patient's datasets that meet the technical criteria from the VSS and can print related reports to serve health insurance reimbursement purpose.
Printing invoices and financial reports	Able to print invoices customized to the hospital's characteristics.
	Able to print various types hospital cost reports (for custom service), including examination fee report, advance fee report, total cost report.
	Able to summarize and print financial reports of hospital costs and health insurance.

Human resource and salary management module

This module features three main functionalities as listed below.

Human resource management	Manage staff's personal information
	Manage training and education programs
	Manage contracts
	Able to search details of each staff bio (career, education, family background, abroad trips)
	Statistics and reporting: Generate and print human resource reports under standardized formats from the MoH and customized reports requested from the hospital. For instance, demographics summaries, salary summaries.
Salary and social insurance management	Manage salary scales and incentives of each staff
	Social insurance management
	Timekeeping system: by month, overtime, surgery, shifts
Statistics and reporting	Manage and print reports requested from the state finance agency such as salary reports and timekeeping reports

	Common templates and reports for human resource and salary management system

Supervision module

This module is to manage the supervised facilities. Main functionalities include:

- Managing training and education programs of supervised facilities.
- Managing primary care and routine health examination activities from lower-level facilities (if available).
- Managing health programs (if applicable).
- Managing supervision reports.

Medical device management module

The main functionalities of this module is given as followed.

Manage medical devices with a standardized nomenclature system	Build a standardized medical device nomenclature system that is used for the whole hospital.
	Have an inventory management functionality for medical devices.
	Have a functionality that allows creating, editing, and removing items in the standardized list.
Managing medical device usage status	Able to manage current usage status of the medical devices and generate related reports.
Managing new devices	Have a data entry functionality for receipt of new devices with specifications for devices received from the providers and devices received from the hospital's wards.
Managing device distribution	Have functionality to manage device distribution in the hospital, and device movement between the wards and to other hospitals.
Managing device maintenance	Have functionality to manage device fixing and maintenance.
Managing device upgrade	Able to identify the technical upgrades that increase device's value.
Managing device liquidation	Have liquidation management functionality.
Module upgrade capability	Able to add functionality items to meet the latest regulations from the MoH and the DoH.
Calculation and searching	Able to automate some calculation tasks and provide advanced searching functions based to multiple criteria.
Reporting	Able to generate and print reports under the standardized templates from the MoH.
	Able to create reports customized to DoH's and hospital's requests.

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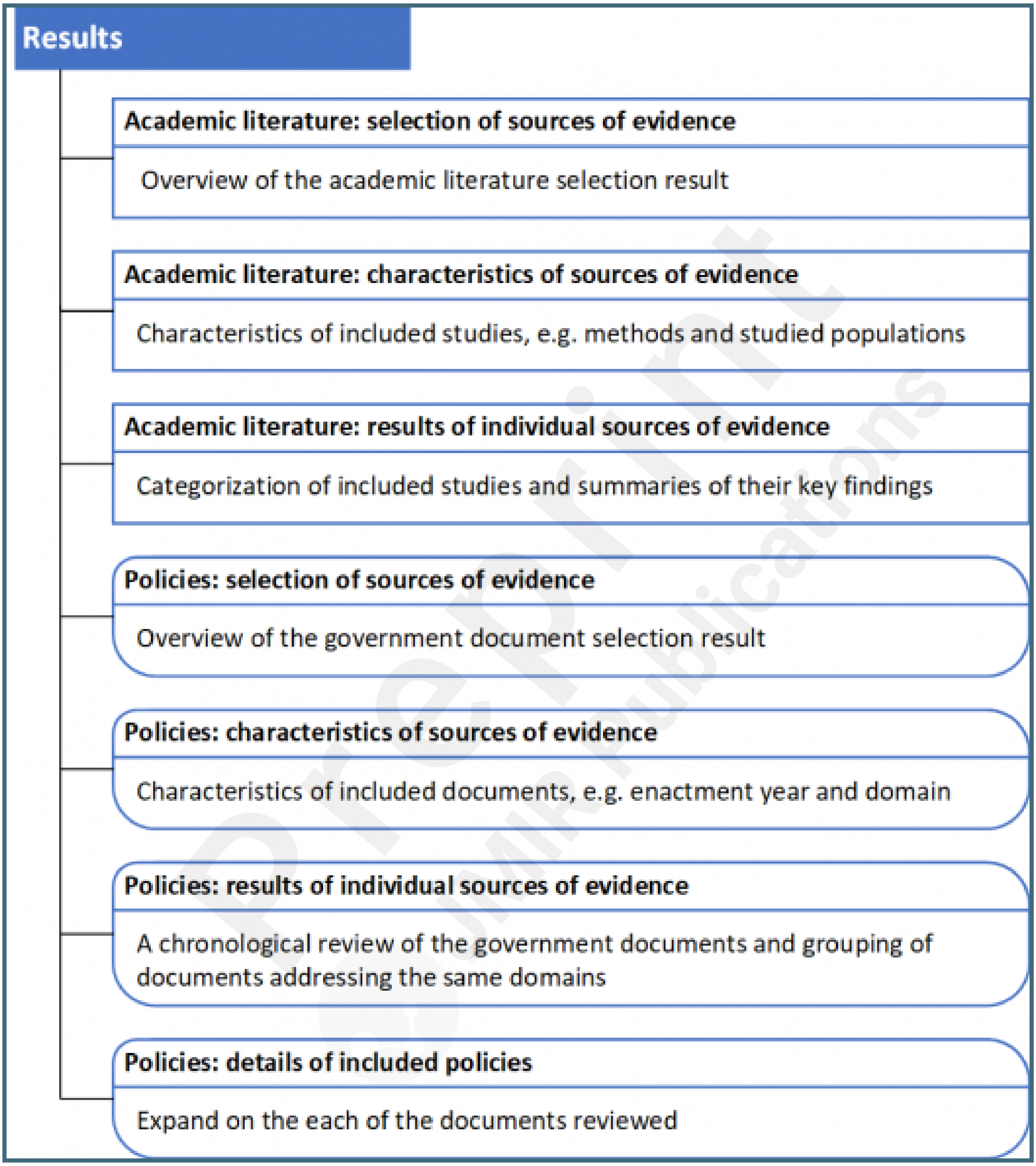
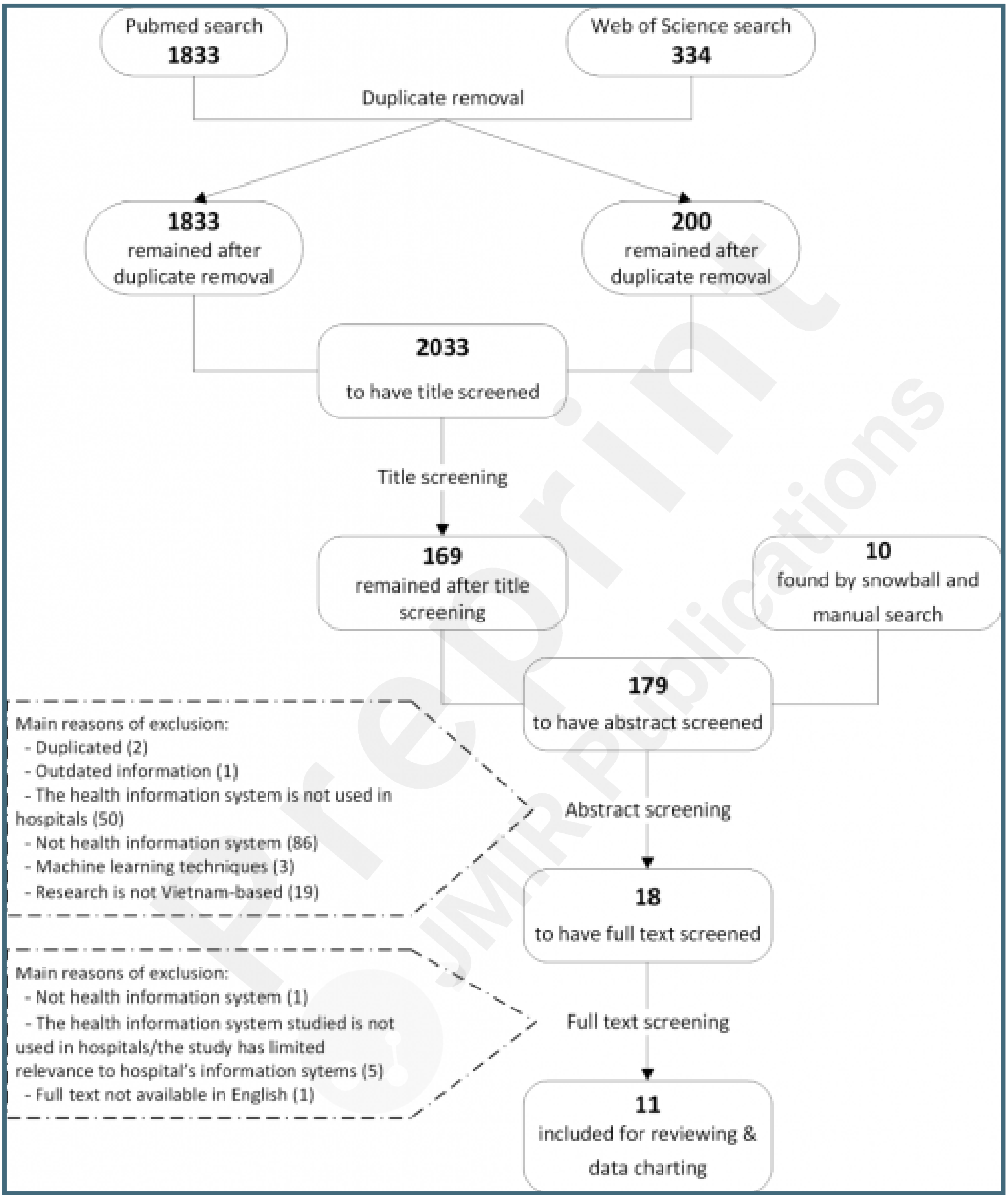
Supplementary Files

Figures

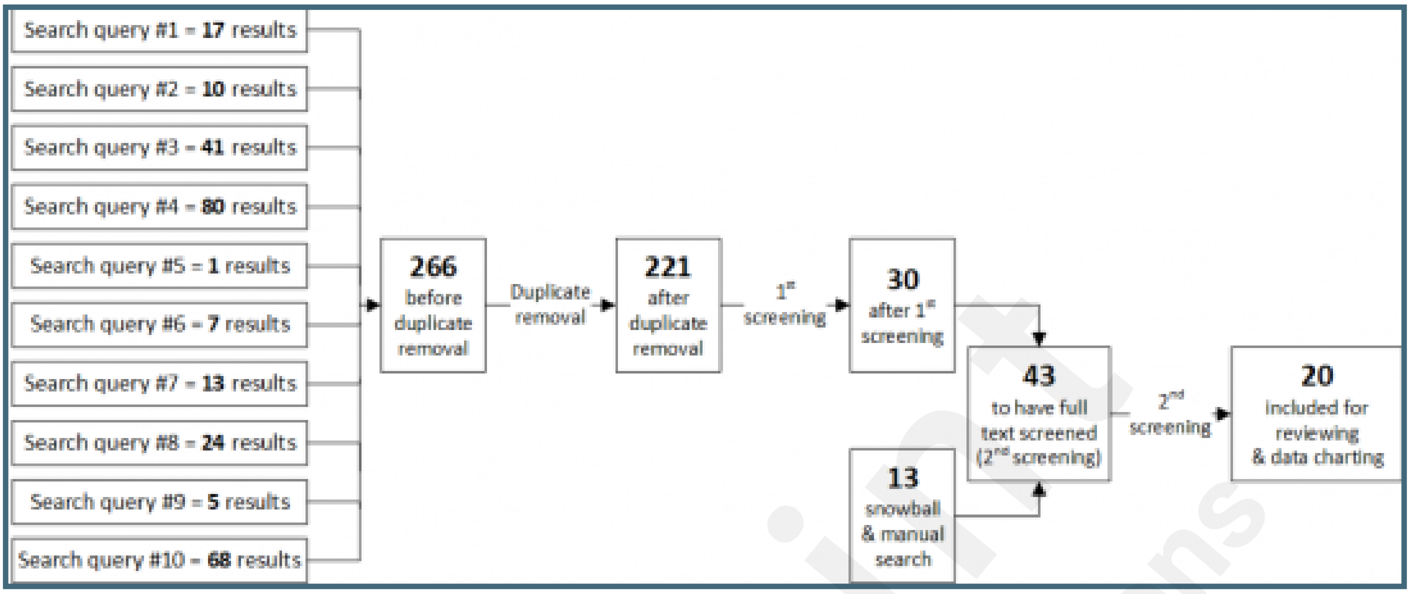
Layout of the Results section.



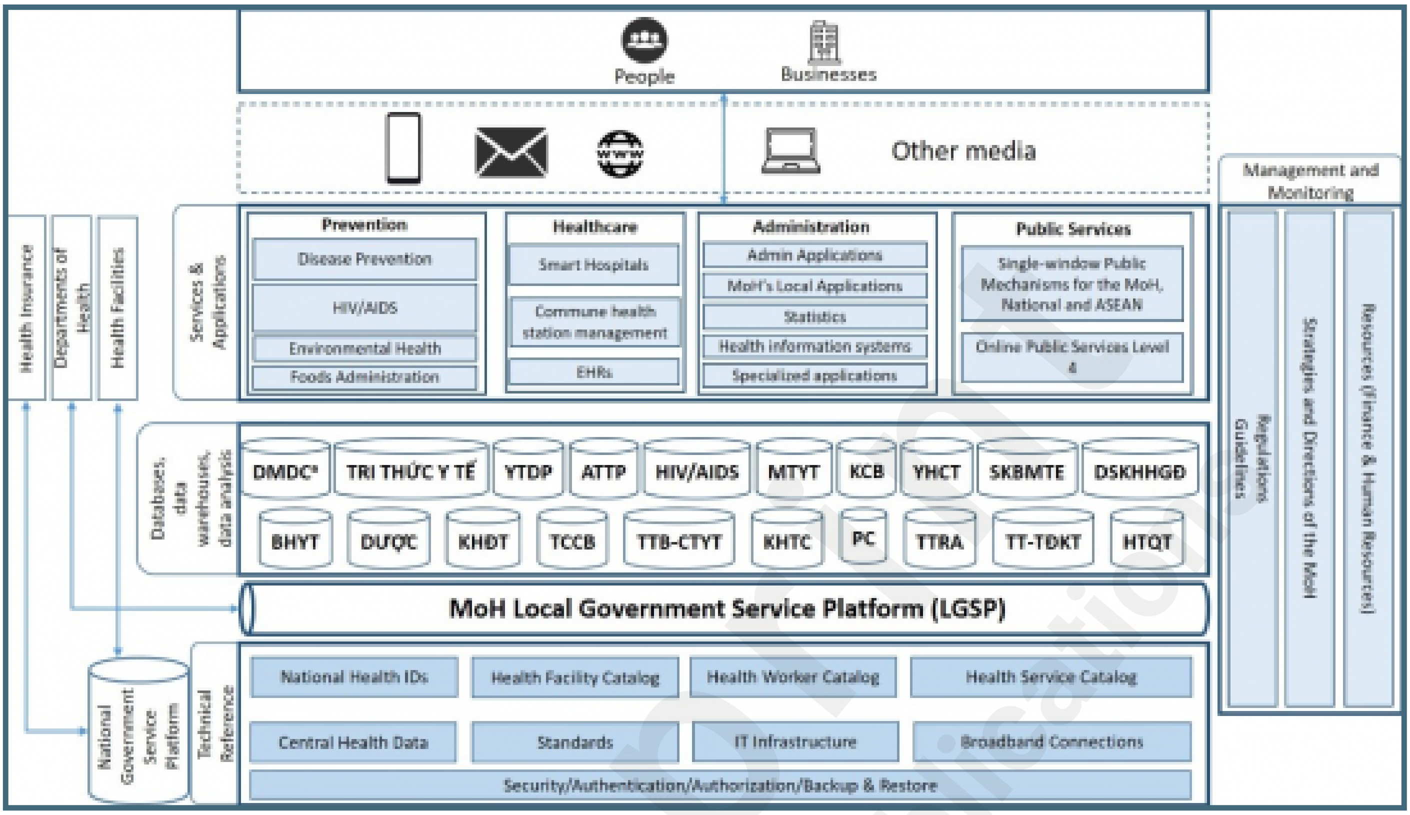
Summary of academic publication selection.



Summary of government document selection.



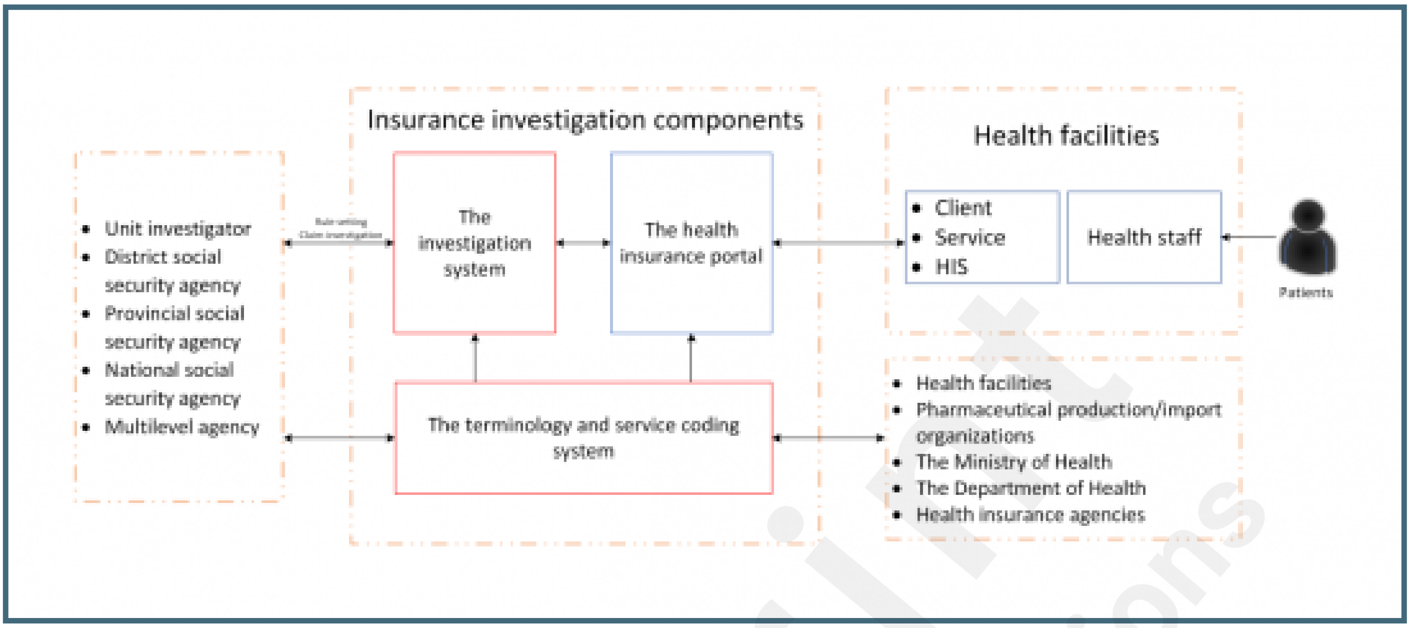
The MoH's eHealth Architecture Framework version 2.0. Adapted from Vietnam Ministry of Health[52].



Levels of HIT Maturity.

	IT infrastructure	HIS	LIS	RIS-PACS	EMR	Administration and Operation Software	Security and Information Safety	Non-functional Criteria	Extra capabilities
HIT Level 7	Level 7	Level 7	Advanced	Advanced	Advanced	Advanced	Advanced	Advanced	- "Paperless" hospital if all relevent criteria are met - CDSS level 3 supporting doctors' decisions related to treatment protocols and treatment results using suitably customized templates - Data in CDR is analysed to improve care quality, patient safety, and care efficiency - Clinical data can be readily shared for stakeholders in care coo rdination based on HL7 standards - Continuous reports of hospital services using the data collected
HIT Level 6 (Smart hospital)	Level 6	Level 6	Advanced	Advanced	Basic	Advanced	Advanced	Advanced	- CDSS level 2 providing: evidence-based warnings for treatment i.e. health and medicine advices, drug information/interaction check, and initial order/prescription violation indentification rules - All structured forms i.e. progress notes, consultation notes, problem lists, and discharge summaries are digitalized - Closed-loop management of drugs, using identification technologies to assist drug administration
HIT Level 5	Level 5	Level 5	Advanced	Advanced	n/a	Basic	Basic	Basic	PACS can replace physical films
HIT Level 4	Level 4	Level 4	Advanced	Basic	n/a	Basic	Basic	Basic	- PACS allows doctors to access images outside the imaging department - Electronic ordering - Electronic management of inpatient orders
HIT Level 3	Level 3	Level 3	Basic	n/a	n/a	Basic	Basic	Basic	- Electronic records having digital vital sign records, nursing notes, medical procedures, and surgical procedures are stored in CDR - CDSS level 1 assisting electronic prescription (new or historic prescription) - Pharmacy information available in the hospital network and supported with CDSS
HIT Level 2	Level 2	Level 2	n/a	n/a	n/a	n/a	n/a	n/a	- A CDR consisting the nomenclature and coding systems, pharmacy, orders, and test results (if available) - Data in CDR can be shared between stakeholders for care coordination
HIT Level 1	Level 1	Level 1	n/a	n/a	n/a	n/a	n/a	n/a	Patient's information can be accessed electronically

The VSS health insurance and investigation structure.



Multimedia Appendixes

Literature search strategies.
URL: http://asset.jmir.pub/assets/fbebbe502c5b31207681c290dfe45f57.doc

Circular 46/2018/TT-BYT on Regulations for electronic medical records.
URL: http://asset.jmir.pub/assets/fb4bf7155960ae72b9bbbd81811abd4e.doc

Circular 53/2014/TT-BYT on Required conditions for provision of health IT activities.
URL: http://asset.jmir.pub/assets/f3d856da8bc3da1652ecdf248756fe46.doc

Circular 39/2017/TT-BTTTT on Technical standards for IT implementation in state organizations.
URL: http://asset.jmir.pub/assets/e4d2e2a8e72bd5caa6385e0378f1ce31.doc

Circular 48/2017/TT-BYT on Regulations on data exchange in management and reimbursement of health insurance claims.
URL: http://asset.jmir.pub/assets/277be853d715045d669b15afe338a92d.doc

Decision 4159/QD-BYT year 2014 on Guidance on ensuring security of electronic health data in health organizations.
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URL: http://asset.jmir.pub/assets/811e1289cd650f1598fc8e36d6a2426a.doc

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URL: http://asset.jmir.pub/assets/12dd62811d0d99ef864519964d106304.doc